

Analysis of Vibrations in Kinematic Chain of Twin Drive Systems of Rotary Kiln in Continuous Process Industries

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Abstract

The RC04 twin-drive rotary kiln at UltraTech Cement’s plant in India has experienced speed-dependent vibrations since commissioning in 2014, with amplitudes reaching 4.5 mm/s within a critical band of 2.5–3.2 rpm - a range that sits within the kiln’s normal operating window of 2–5 rpm. Component replacements, including a full girth gear overhaul in 2023, have not resolved the problem. This project develops a finite element model of the RC04 drivetrain to identify the resonating component. Gear mesh stiffness is calculated using the ISO 6336 semi-empirical method in MATLAB (Otto, Fromberger, and Stahl 2024) and validated against Abaqus modal results to within 2.3%. A 3D model built in SolidWorks from engineering drawings is imported into Abaqus for modal analysis. Boundary conditions are established based on the dynamics of granular material flow to correctly represent the equivalent kiln inertia at the output to the Gearbox. With this applied, the dominant torsional mode of the first gearbox stage (21/61) corresponds to a kiln speed of 2.9 rpm - within the observed critical band. The torsion shaft contributes no critical modes within the operating range. Mitigation options are then evaluated within the constraints of the existing housing and foundations. Idler gear configurations are assessed one by one; the solutions that produce sufficient frequency shifts are found to be geometrically infeasible, and those that are geometrically feasible do not shift the critical speed far enough. A technically feasible combined solution is therefore proposed to mitigate vibrations during operation.

Lay Summary

This project investigates a vibration problem in the main rotary kiln of a cement plant, which produces clinker – the key ingredient in cement. The kiln, powered by a twin-drive system, exhibits excessive vibrations at certain operating speeds, which can damage gears, bearings, and couplings, and has previously lead to unplanned shutdowns.

The study validates a numerical theory of the drive train’s dynamic response through computer simulations and proposes a redesign of the kiln’s gear reduction stages to reduce vibration. The solution targets the source of the problem, ensuring the system operates more smoothly.

If implemented successfully, the redesign is expected to extend the life of critical components, reduce unplanned downtime, and improve fuel efficiency. This is particularly important in India, where cement demand is projected to rise from 445 million tonnes in FY24 to 670 million tonnes by 2030 (**Anand2025CementDemand**) . Stabilizing existing kilns ensures they can meet this growing demand, support major infrastructure and housing projects, and minimize the economic and operational costs of avoidable breakdowns.

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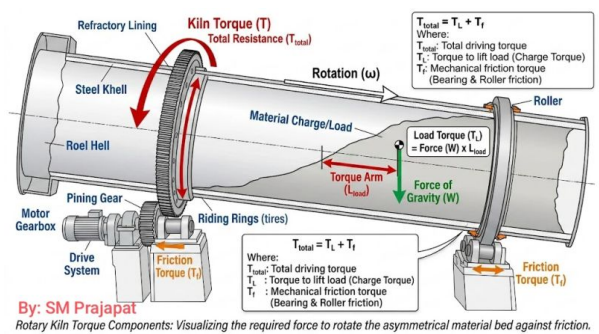
1. Introduction

Machinery plays a crucial role in industrial operations, enabling large-scale production and driving economies across various sectors, especially in continuous process industries like cement manufacturing, where equipment is operated 24/7 for typically 340 days in a year, stopping only for an annual maintenance shutdown. One of the persistent challenges associated with mechanical systems is vibration during operation, which can lead to reduced efficiency, increased wear, and structural failure. As stated by Rao, “any body that has mass and elasticity is capable of oscillatory motion” (Rao 2019). Vance, Zeidan, and Murphy define industrial vibration as arising from inertia forces in reciprocating and rotating components, with amplitude becoming large when excitation frequency approaches a natural frequency of the system: a phenomenon called resonance or critical speed (Vance, Zeidan, and Murphy 2010). Resonance is particularly critical: when the frequency of external forces matches the system’s natural frequency, it leads to excessive vibration. This is commonly observed in rotating machinery, pumps, and large structures like bridges.

In cement manufacturing specifically, large gearboxes driving rotary kilns, ball mills, and crushers operate continuously under heavy loads; hence, even minor deviations in gear mesh, foundation condition, or bearing wear can excite structural resonances and lead to excessive vibration. A pertinent case can be found at UltraTech Cement’s plant in Kalaburagi, India, where resonance is suspected to be the cause of unexplained excessive vibrations within a specific speed range in the drive system of a critical rotary kiln.



(a) Image of a Kiln (Kindle Tech 2026)



(b) Load Diagram (Prajapat 2023)

Figure 1.1

1.1 Rotary Kiln in Continuous Process Industries

The bloom in infrastructure has been possible due to mass production of construction materials like iron and cement. Being two of the most produced materials, their manufacturing origins can be traced back to the rotary kiln. A rotary kiln is a large cylindrical furnace which rotates about its axis, used for high-temperature process manufacturing where chemical reactions take place due to the influence of heat. Its slight slope and slow rotation cause material inside to travel from the feed end to the discharge end, ensuring homogeneous heat distribution and mixing.

Heat is commonly generated by a flame at the discharge end figure 1.2, from the combustion of coal, oil, natural gas, or waste. In order to operate at high temperatures, the inner part of the rotary kiln is lined with refractory bricks, which protect the steel shell, reduce heat loss, and contain the process environment. Brick fall-outs can lead to significant failure, as heat escapes and the affected spot becomes visibly red-hot, necessitating a shutdown for maintenance. Unplanned shutdowns can cause very high production losses and put a plant in a demanding operational situation.

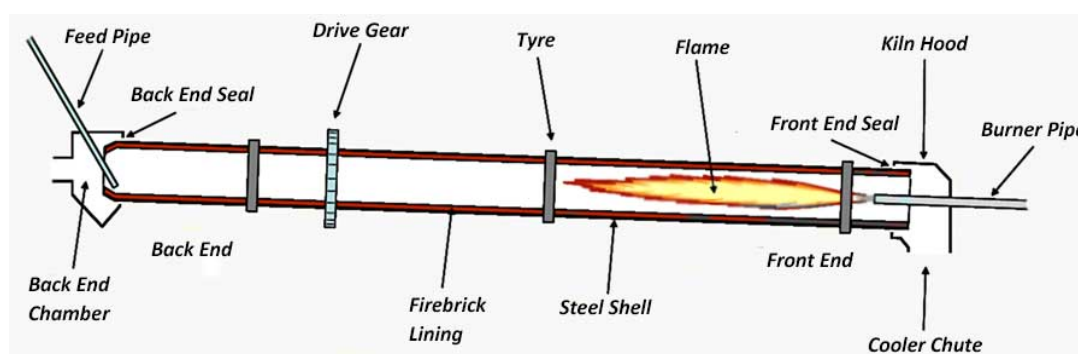


Figure 1.2: Rotary Kiln Structure

For cement manufacturing, the rotary kiln is the heart of the factory. The internal temperature reaches 1400°C to catalyse the chemical reaction of calcination of limestone into clinker, which is further processed to make cement. To drive such a massive system, a kinematic chain delivers high torque to the kiln through speed-reduction stages, allowing continuous rotation. The RC04 kiln at UltraTech Cement's plant in Kalaburagi, India, is a twin drive system, driving the kiln from both sides, with a designed capacity of 12,000 tonnes per day and a shell size of $\varnothing 6\text{ m} \times 88\text{ m}$. Additionally, thick steel tyres fitted tightly to the shell ride freely on support rollers, and two thrust rollers act on the tyres to prevent the shell from sliding down the slope.

1.2 Background of the Problem

RC04 was commissioned in 2014. Almost immediately after commissioning, abnormal vibration behaviour was observed in the kiln drive system. The initial operations indicated high vibrations in the bands of 2.7–3.1rpm, and gearbox foundations were strengthened as a response. The kiln speed is controlled by a variable frequency drive (VFD) and is increased or decreased in steps of 0.2rpm to prevent thermal shock, as the outer shell operates at approximately 700 °C.

A vibration study was conducted in 2014 by the Vibrotech team, with measurements recorded at 2.5, 2.8, 3.0, and 4.8rpm. The relevant key findings were:

- Increased vertical vibration levels at gearbox input shafts — 4.4 mm/s at 2.8 rpm on the pressing side and 3.4 mm/s at 2.9 rpm on the lifting side.
- **Higher vibrations at the first-stage gear mesh frequencies.**

The study recommended avoiding sudden speed increases of more than 0.2rpm during transients and suggested that the manufacturer shift any critical speed by at least 20%. However, due to limited instrumentation and the absence of a structured analytical framework, no conclusive root cause was established, and the vibrational problem persisted.

Within just two years of operation, by 2016, the girth gear and pinions developed cracks at diagonally opposite locations on the gear web and hairline cracks at the gear root. The reasons were initially attributed to design and casting; however, the vibrational problem still persisted after component replacements, suggesting the need to study the vibrational aspect of the problem more thoroughly. Eventually, the girth gear and pinion assemblies were prematurely replaced during the 2023 shutdown.

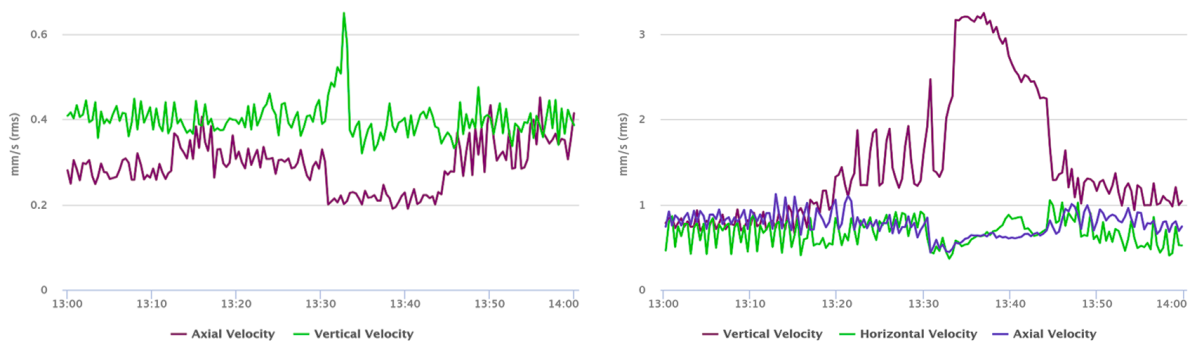


Figure 1.3: Vibration Data

Figure 1.3 shows the high vibrations measured by Gearbox Input Shaft sensors as the kiln speed passes through the problematic speed band.

The problem has since evolved. The critical vibration band has widened from the originally reported 0.4rpm range (2.7–3.1rpm) to approximately 0.7rpm, now spanning 2.5–3.2rpm. Cracks have also appeared in the top layer of the gearbox foundations, suggesting that boundary conditions may have changed over time, potentially shifting natural frequencies and widening the resonance band. Outside this critical range, the system exhibits stable behaviour, with vibration amplitudes below 1mm/s and oil analysis confirming healthy drivetrain condition. This contrast strongly reinforces the hypothesis that the issue is not due to general mechanical degradation, but rather a speed-dependent dynamic phenomenon — resonance.

1.3 Operational Challenges

The existence of a critical speed band within the normal operating range of the kiln creates significant operational constraints. The kiln’s designed operating range is 2–5rpm, and the critical band of 2.5–3.2rpm sits squarely within this range, effectively restricting continuous operation to below 2.5rpm or above 3.2rpm.

To manage this, operators must skip the critical band entirely by commanding a single speed step of 0.7rpm under load, rather than the standard increments of 0.2rpm. This puts the outer shell — which operates at approximately 700°C — at risk of thermal shock. Even with this precaution, the kiln still passes through the critical band in approximately two minutes as the VFD ramps to its set speed. During this transient, vibration spikes reaching 4.5mm/s are consistently observed in the vertical and horizontal directions at both gearboxes and pinion bearings, compared to a normal level of below 1mm/s.

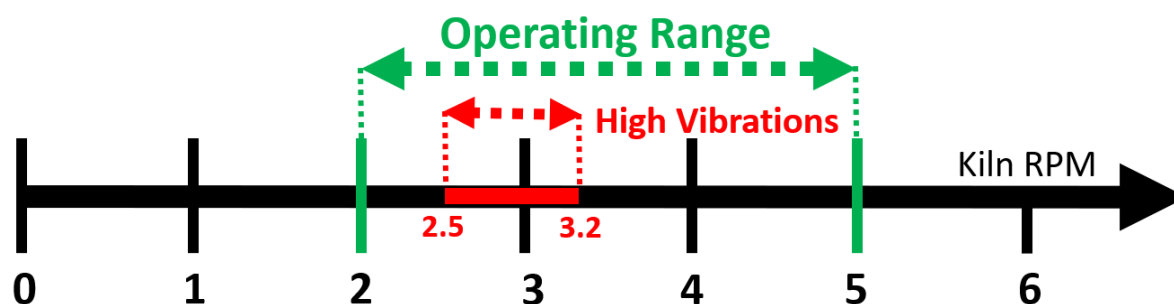


Figure 1.4: Problematic Operational Band

Beyond the risk to the drive components themselves, any change in kiln speed has process-level consequences. Speed changes cause temporary shifts in the burning zone, affecting kiln inlet temperature and burning efficiency, which in turn require adjustments to

fuel rate and induced draft fan speed. Reduced material mixing, decreased heat transfer efficiency, and increased risk of material buildup on the kiln shell all impact product quality and kiln stability. The repeated forced transients through the critical band therefore introduce not only mechanical risk but also process instability, making this a bottleneck for plant operations.

1.4 Gap in Understanding and Scope of This Work

The 2014 Vibrotech study established that elevated vibrations exist within specific speed bands and identified gear mesh frequency as a likely excitation source. However, it stopped short of identifying which component’s natural frequency is being excited, or why. The root cause remains unresolved a decade later.

The subsequent mechanical failures, the widening of the critical band, and the foundation cracking all suggest that the system dynamics have evolved, making the original findings even less reliable as a basis for corrective action. What is needed is a structured analytical investigation that identifies the resonating component and quantifies the conditions under which resonance occurs.

This work addresses that gap through a combined modelling, simulation, and validation approach. The gear tooth stiffness of the drive system is mathematically modelled based on ISO 6336 to theoretically calculate the gear mesh critical frequency. A 3D model of the drive system is constructed in SolidWorks from engineering drawings, and modal analysis is performed using **Finite Element Analysis** in Abaqus to extract natural frequencies and vibrational mode shapes. Using the results, design strategies are suggested to mitigate vibrations.

1.5 Aim and Objectives

Aim

The aim of this project is to develop and validate a practical workflow for predicting and mitigating gear–mesh resonances in an industrial gearbox, by combining ISO 6336–based mesh–stiffness calculations, finite element modal analysis, and measured vibration data. This supports industry in diagnosing root causes of excessive vibration and designing targeted, evidence–based mitigation measures at minimal cost and downtime.

Objectives

To achieve this aim, the project is structured around the following specific objectives:

1. Characterise the problem and operating context

- (a) Compile the gearbox layout, duty cycle, and operating speed/torque envelope from engineering documentation and vibration reports.
- (b) Identify dominant mesh frequencies and existing critical-speed concerns from measured spectra.

2. Model development

- (a) Implement a MATLAB tool that estimates single-pair dynamic mesh frequencies for the subject gear pairs using semi-empirical ISO 6336 methods, providing theoretical critical speed predictions.
- (b) Create a 3D CAD assembly of the drive system (gears, shafts, bearings, housing) in SolidWorks from engineering drawings, suitable for downstream FEA.

3. Numerical analysis

- (a) Perform linear modal analysis in Abaqus on the assembled model, under realistic support and boundary conditions, to obtain natural frequencies and mode shapes over the relevant operating range.
- (b) Compute frequency response functions (FRFs) at measurement locations to identify which modes contribute most to vibration at gear-mesh harmonics.

4. Validation and diagnosis

- (a) Compare ISO-based critical speed estimates and FE modal results with plant vibration data, and confirm whether predicted resonances occur within $\pm 10\%$ of operating mesh frequencies.
- (b) Use mode-shape and FRF information to identify the components (gears, shafts, bearings, housing) that dominate each resonant response and thus constitute the primary vibration sources.

5. Mitigation strategy and reassessment

- (a) Propose at least one case-specific vibration-mitigation measure (e.g. modifying stiffness or mass distribution, adjusting supports/foundation, or introducing damping elements) based on the diagnosed resonant modes.
- (b) Incorporate the proposed changes and demonstrate that the relevant critical frequencies are shifted outside the operating band and that peak response amplitudes are reduced by at least 50% relative to the baseline.

2. Literature Review

This chapter discusses existing literature surrounding resonance behaviour in gear transmission systems. It focuses on how gear mesh excitation, structural dynamics, and system modelling approaches contribute to vibration and resonance in geared systems. The review begins with methods for modelling gear mesh stiffness, followed by an overview of modal analysis techniques used in rotating machinery, and concludes with considerations of boundary conditions and dynamic modelling principles relevant to gear system analysis.

2.1 Modal Analysis in Rotating Machinery

Modal analysis is a fundamental tool for understanding the vibration behaviour of geared transmissions and rotating machinery.(Han et al. [2017](#); Herisanu and Marinca [2021](#)) It identifies the natural frequencies, mode shapes, and damping characteristics of a system, and predicts how the structure responds to dynamic excitations such as gear mesh forces.(Kuemmlee, Gross, and Kolerus [2013](#))

2.1.1 Definition and Purpose of Modal Analysis

Modal analysis represents the vibration of a complex mechanical system as a combination of simple oscillatory modes.(Herisanu and Marinca [2021](#)) Each mode is defined by:

- a natural frequency f_n at which the system tends to vibrate,
- a mode shape ϕ_n describing the deformation pattern,
- a damping ratio ζ_n describing how quickly the vibration dies out.

For gear transmissions, modal analysis is essential for:

- critical speeds where mesh harmonics coincide with natural frequencies,(Han et al. [2017](#))
- identifying modes that efficiently radiate noise,(Jiang et al. [2023](#))
- estimating vibrations' influence tooth stress and bearing loads,(Kong et al. [2023](#))
- optimising design by adjusting stiffness, mass distribution, and bearing properties.(Ouyang et al. [2019](#))

2.1.2 Natural Frequency Calculation: Lumped-Mass and FE Methods

Natural frequencies come from solving the standard eigenvalue problem

$$K\phi = \lambda M\phi, \quad (2.1)$$

where K is the stiffness matrix, M is the mass (or inertia) matrix, ϕ is the mode shape, and $\lambda = \omega_n^2$ is the eigenvalue.(Han et al. 2017)

This equation expresses a simple idea: *in a free vibration mode, the restoring forces (contained in K) must balance the inertia forces (contained in M).* Solving this matrix equation produces a set of natural frequencies and their associated shapes.(Mobley 1999)

Two common approaches are used:

- **Lumped-mass models**, useful for shafts, gears, and simplified rotordynamics,(Han et al. 2017)
- **Finite element models**, used for housings, carriers, and flexible shafts.(Kong et al. 2023)

Experimental Modal Analysis (EMA) validates these models using impact-hammer tests and accelerometers.(Kuemmlee, Gross, and Kolerus 2013) In published studies, differences between measured and FEA frequencies are typically below 6%, confirming that well-constructed models accurately reproduce real gearbox behaviour.(Ouyang et al. 2019)

2.1.3 Modal Analysis in Planetary and Multistage Systems

Planetary and multistage transmissions involve many interacting components, producing a rich modal spectrum.(Han et al. 2017; X. Yang et al. 2023)

Modal energy distribution indicates which component (planet, sun, carrier, shaft) dominates a mode and therefore which part of the gearbox will respond most strongly to excitation.(Jiang et al. 2023)

2.1.4 Mode Shapes and Modal Participation

Different families of mode shapes are typically observed in gear transmissions: global housing modes, shaft bending modes, torsional/rotational modes, and local panel modes of the housing.(Han et al. 2017; Kong et al. 2023) Coupled lateral-torsional or axial-torsional modes are particularly important in helical and double-helical systems, where oblique mesh forces couple multiple degrees of freedom.(Chen et al. 2016)

Transient or forced vibrations can be expressed using modal superposition:

$$x(t) = \sum_{n=1}^N \phi_n q_n(t), \quad (2.2)$$

where each modal coordinate $q_n(t)$ behaves like an independent single-degree-of-freedom oscillator. (Han et al. 2017) This greatly simplifies response calculations and allows one to study how individual modes contribute to the overall vibration. (Herisanu and Marinca 2021)

2.1.5 Campbell Diagrams and Critical Speeds

A Campbell diagram overlays natural frequencies with speed-dependent excitation frequencies such as mesh harmonics. (Han et al. 2017) Critical speeds occur when:

$$f_{\text{mesh}} = n f_n, \quad (2.3)$$

where n is the harmonic order. (Chen et al. 2016) Intersections between mesh lines and modal lines identify potential resonances in the operating range. Such diagrams are widely used to detect problematic modes early in the design stage. (Król 2021)

2.1.6 Frequency Veering and Mode Coupling

As shaft speed increases, some natural frequency curves approach one another, change slope, and then separate—a phenomenon known as *frequency veering*. (Chen et al. 2016) When veering occurs, the modes effectively exchange characteristics. In gear systems, this is caused by:

- gyroscopic effects, (Chen et al. 2016)
- speed-dependent mesh stiffness, (X. Yang et al. 2023)
- shaft geometric effects. (Król 2021)

Frequency veering can complicate dynamic predictions and often requires nonlinear or speed-dependent models. (Li et al. 2025)

2.1.7 Modal Damping and Quality Factors

The damping ratio ζ_n determines the sharpness of resonance. The quality factor is

$$Q_n = \frac{1}{2\zeta_n}. \quad (2.4)$$

Typical values in gear transmissions are: (Han et al. 2017)

- low damping ($\zeta < 0.02$): sharp, high-amplitude resonances,
- moderate damping ($\zeta \approx 0.05$): manageable response,
- high damping ($\zeta > 0.10$): broad, lower resonant amplification.

Friction-based devices such as ring dampers can greatly increase damping, reducing tooth stresses and radiated noise, especially in thin-walled gears.(Xu et al. 2023; Shweiki et al. 2019)

2.1.8 Connection Between Modal Analysis and Gear Mesh Excitation

Each mode is excited according to its modal force

$$F_n = \phi_n^T F_{\text{mesh}}(t), \quad (2.5)$$

where $F_{\text{mesh}}(t)$ includes the time-varying mesh stiffness and transmission-error forces.(X. Yang et al. 2023; Abruzzo et al. 2023) Modes with large $|F_n|$ respond strongly, while modes with very small $|F_n|$ are effectively decoupled from the mesh.(Mélot et al. 2022)

2.1.9 Load-Dependent Modal Shifts

Transmission stiffness changes with load, especially under high torque and in lightweight or thin-walled designs.(Kong et al. 2023; Shweiki et al. 2019) As a result:

- natural frequencies may shift by several hertz,(X. Yang et al. 2023)
- mode ordering can change,(Jiang et al. 2023)
- high-load or overload conditions (e.g., double or triple tooth contact) modify the stiffness landscape.(Celikay, Donmez, and Kahraman 2021)

This means that the critical speed at rated load may differ from that at low load. Accurate design therefore requires analysing a range of operating conditions.(Li et al. 2025)

2.1.10 Application: Modal Analysis for Noise Prediction

Gear noise is primarily radiated by the gearbox housing. The acoustic power of each mode scales with

$$P_n \propto |v_n|^2 S_n, \quad (2.6)$$

where v_n is the vibration velocity of the housing and S_n is the radiating surface area.(Han et al. 2017) Modes involving large housing deformation (such as panel or swing modes)

typically dominate airborne noise, even if the internal gear motion is small.(Jiang et al. 2023) Modal energy analysis therefore plays a key role in identifying which modes are the most acoustically significant.(Herisanu and Marinca 2021)

2.1.11 The role of FEA Softwares: Abaqus

Computer simulations play a central role in modern gear dynamics and modal analysis. Although analytical and lumped-parameter models provide valuable insights, real gearboxes contain complex geometries, flexible housings, non-uniform shaft sections, and detailed contact interfaces that cannot be captured accurately with simplified models alone. Finite Element Analysis (FEA) therefore becomes essential for predicting vibration behaviour with engineering accuracy.

2.2 Boundary Conditions

When modelling gears, shafts, and housings in dynamic analysis, two aspects are crucial: (1) the *boundary conditions* that define how the system is supported, loaded, and constrained, and (2) the way *d'Alembert's principle* allows these rotating loads and constraints to be included naturally into the equations of motion.

Boundary Conditions in a Rotating Gear System

Boundary conditions specify how each part of the gearbox interacts with its environment and therefore strongly influence the computed natural frequencies and mode shapes.(Stanis et al. 2020; Anonymous 2020) For a practical transmission, key boundary conditions include:

- the mounting and flexibility of the gearbox housing (rigid foundation, elastic mounts, or frame flexibilities),
- bearing translational and rotational stiffnesses at each shaft journal,
- applied input torque and any speed-dependent drive characteristics,
- reaction torque and constraints at the output (e.g. generator, conveyor, or kiln),
- axial and radial constraints from couplings,
- assembly preload in bearings and joints.

Experimental and numerical studies on transmission housings and geared rotors show that changing support conditions or bearing stiffness can shift certain natural frequencies by tens of percent and can even change which components dominate a mode shape.(Stanis

et al. 2020; Han et al. 2017) For example, stiffer bearing supports generally push bending-dominated shaft modes to higher frequencies, whereas softer couplings and foundations lower torsional modes and can introduce additional low-frequency rigid-body-like modes.(Han et al. 2017; Anonymous 2020) Similarly, preload or fit conditions that increase local contact stiffness at bearings or gear meshes can shift mesh-related modes and modify how vibration energy is transmitted into the housing.(Marafona et al. 2021)

Using d’Alembert’s Principle to Incorporate Rotational Loads

d’Alembert’s principle states that a dynamic system can be analysed as if it were in static equilibrium by adding inertial forces and moments that oppose the accelerations. For a rotating shaft or gear, this leads to equivalent body forces and torques that represent angular acceleration, centrifugal effects, and gyroscopic coupling.(NPTEL 2010) Classical rotordynamics studies derive the governing equations of laterally vibrating rotating shafts in cylindrical coordinates by applying d’Alembert’s principle: the centrifugal body force produces an effective axial stress and spin-stiffening (or softening), and the mass moment of inertia generates gyroscopic moments that couple orthogonal bending directions.(Behzad and Bastami 2004; Khader 1995)

More recently, d’Alembert’s principle has been combined with finite element models of complex structures: modal masses obtained from FE are used to define inertial loads, which are then applied in static analyses to recover stress fields under dynamic excitation(Sánchez-Sarmiento and Sorichetti 2022). The same idea can be extended to geared rotor-bearing systems by treating:

- inertial torques opposing shaft angular accelerations,
- centrifugal body forces acting on rotating gears and shafts,
- gyroscopic moments arising from discs and gears spinning about their polar axes,

as additional loads or stiffness terms in the equilibrium equations.

In the context of this gearbox, d’Alembert’s principle allows operating rotational effects to be embedded directly into the modal analysis. Instead of computing modes for an idealised non-rotating structure, the FE model is solved with centrifugal loads, gyroscopic coupling and torque-dependent boundary conditions included, so that the resulting natural frequencies and mode shapes reflect the actual running condition of the gearbox.(Behzad and Bastami 2004; Marafona et al. 2021) This improves agreement with measured vibration spectra, particularly for high-speed shafts where spin-softening and gyroscopic splitting make the rotating natural frequencies deviate significantly from their stationary counterparts.(Behzad and Bastami 2004; Sánchez-Sarmiento and Sorichetti 2022)

2.3 Gear Mesh Stiffness Modelling

Gear mesh excitation originates from time-varying mesh stiffness as tooth pairs alternately come in contact, producing excitation at the tooth engagement frequency (gear mesh frequency, GMF) and its harmonics. ISO 6336 defines an approximate semi-empirical formula for mean mesh stiffness based on Weber-Banaschek's analytical tooth deflection model (Marafona et al. 2021). Experimental validation for a spur-gear mesh yielded 4880 rpm measured vs 5030 rpm calculated (4% error), confirming the standard's adequacy for estimations (Otto, Fromberger, and Stahl 2024).

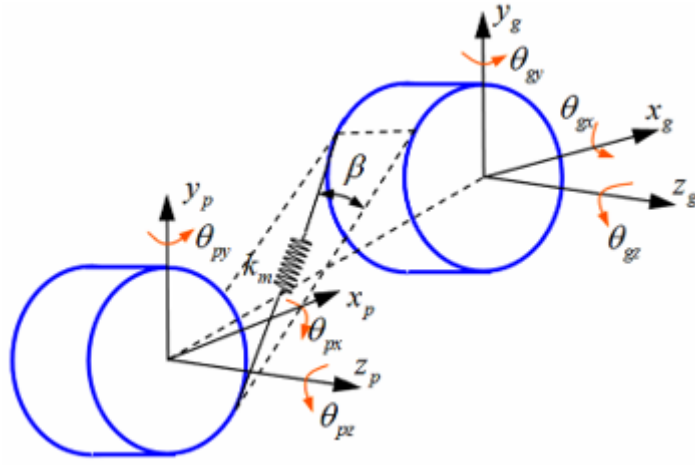


Figure 2.1: Sketch of Gear Mesh Model (Chen et al. 2016)

More sophisticated analytical approaches include the potential-energy method, which sums tooth bending, shear, axial compression, and Hertzian contact contributions and has been extended to include tip relief, cracks and spalling (Abruzzo et al. 2023). These models allow us to theoretically calculate critical frequency of gear pairs occurring in the drive.

2.3.1 ISO 6336 Semi-Empirical Method for Mesh Stiffness

The ISO 6336 standard calculates mean mesh stiffness using the Weber–Banaschek theory. Total deflection is decomposed into three additive components:

$$\delta_{\text{total}} = \delta_{\text{contact}} + \delta_{\text{tooth}} + \delta_{\text{body}}. \quad (2.7)$$

(Otto, Fromberger, and Stahl 2024) Mesh stiffness k is defined as the resistance of the gear pair to deflection caused by applied load along the line of action:

$$k = \frac{F}{\delta} \quad (2.8)$$

where F is the normal load and δ is the relative deflection.

Torsional mesh stiffness relates to linear stiffness via:

$$k_{\text{torsional}} = k_{\text{linear}} r_b^2 \quad (2.9)$$

where r_b is the gear base radius.

Transmission error (TE) is the difference between the actual angular position of the driven gear and its ideal position for perfectly rigid, conjugate gears:

$$\text{TE} = \frac{\delta}{r_{b,2}} - \frac{\delta}{r_{b,1}} \quad (2.10)$$

The gear radii follow from

$$r_i = \frac{z_i m}{2}, \quad i = 1, 2, \quad (2.11)$$

and each gear is approximated as a solid disk with

$$m_i = \rho \pi r_i^2 b, \quad I_i = \frac{1}{2} m_i r_i^2. \quad (2.12)$$

ISO 6336 provides a semi-empirical mesh-stiffness expression based on geometry, profile shift, and pressure angle. Using coefficients $C_1 \dots C_9$, a parameter q is computed and converted into a theoretical stiffness.

$$q' = C_1 + \frac{C_2}{z_{n1}} + \frac{C_3}{z_{n2}} + C_4 x_1 + \frac{C_5 x_1}{z_{n1}} + C_6 x_2 + \frac{C_7 x_2}{z_{n2}} + C_8 x_1^2 + C_9 x_2^2 \quad (2.13)$$

C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
0.04723	0.15551	0.25791	-0.00635	-0.11654	-0.00193	-0.24188	0.00529	0.00182

Table 2.1: Constants for Eq. 2.13

$$c_{\text{th}} = \frac{10^{-3}}{q}. \quad (2.14)$$

This stiffness is modified by ISO correction factors for material, body flexibility, and tooth bending:

$$c = c_{\text{th}} C_m C_r C_b \cos \beta. \quad (2.15)$$

The dynamic model reduces the gear inertias to the mesh line of action using

$$I_{\text{red}} = \frac{I_1 I_2}{I_1 + I_2} (r_1^{-2} + r_2^{-2})^{-1}. \quad (2.16)$$

The torsional natural frequency is then

$$f = \frac{1}{2\pi} \sqrt{\frac{c}{I_{\text{red}}}}, \quad (2.17)$$

which is converted into resonance speeds for each gear:

$$n_i = \frac{30000}{\pi} \frac{\sqrt{c/I_{\text{red}}}}{z_i}, \quad i = 1, 2. \quad (2.18)$$

These speeds represent the operating conditions at which the gear mesh exhibits maximum dynamic amplification and should generally be avoided.

2.3.2 Potential Energy Method (PEM)

A more rigorous method, based on the strain energy stored in the deformed gear tooth and body. (H. Yang et al. 2022)

Hertzian Contact Deformation

Local elastic deformation at the tooth contact point (Hertzian deformation):

$$\delta_{\text{contact}} = \frac{10.92}{E'} \left(3.1 + \frac{1}{0.294} \right) \frac{F}{b} \quad (2.19)$$

where E' is the reduced modulus of elasticity, F is load per unit width, and b is the Hertzian contact width. (Zhang et al. 2017)

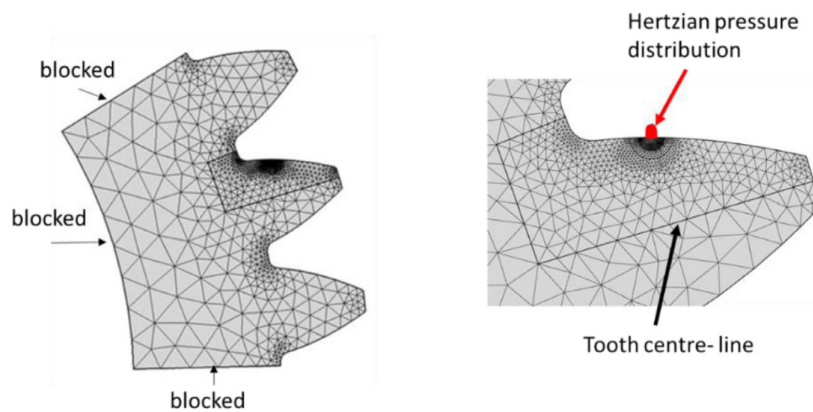


Figure 2.2: Hertzian Contact Model (Sainsot and Vexlex 2020)

Strain Energy Decomposition

Total potential energy:

$$U_{\text{total}} = U_{\text{bending}} + U_{\text{shear}} + U_{\text{axial}} + U_{\text{Hertzian}} \quad (2.20)$$

Bending energy:

$$U_b = \int_{z_{\text{root}}}^{z_{\text{load}}} \frac{M(z)^2}{2EI} dz \quad (2.21)$$

Shear energy:

$$U_s = \int_{z_{\text{root}}}^{z_{\text{load}}} \frac{1.2 V^2}{2GA} dz \quad (2.22)$$

where V is the shear force and A the cross-sectional area.

Axial compression energy:

$$U_a = \int_{z_{\text{root}}}^{z_{\text{load}}} \frac{N^2}{2EA} dz \quad (2.23)$$

where:

$$N = F \sin(\text{pressure angle}) \quad (2.24)$$

Hertzian contact energy:

$$U_{\text{Hertz}} = \frac{1}{2} k_{\text{Hertz}} \delta_{\text{contact}}^2 \quad (2.25)$$

Hertzian contact stiffness (Sainsot–Valex formulation):

$$\frac{1}{k_H} = \frac{\pi}{4E'} \left[\ln \left(\frac{1 + \sqrt{1 - 2d}}{1 - \sqrt{1 - 2d}} \right) - 0.429(1 - \sqrt{1 - 2d}) + \frac{\sqrt{1 - 2d}}{2d} \right] \quad (2.26)$$

where d is the normalized datum depth. (Sainsot and Valex 2020)

Single-Tooth Stiffness from Potential Energy

By complementary work:

$$k_{\text{tooth}} = \frac{F^2}{2U_{\text{total}}}. \quad (2.27)$$

Substituting and integrating along the tooth profile yields the single-tooth stiffness as a function of load position. The main resonance frequency of a gear stage is:

$$f_{\text{resonance}} = \frac{1}{2\pi} \sqrt{\frac{k_{\text{mean}}}{I_{\text{red}}}} \quad (2.28)$$

3. Methodology

3.1 Modelling

3.1.1 Mathematically Model Gear Teeth Stiffness in MATLAB

The first step in the methodology is to obtain theoretical estimates of the gear-mesh critical frequencies for each reduction stage in the kiln drive. This is achieved by implementing the ISO 6336 semi-empirical method for mean mesh stiffness in MATLAB, using the gear geometry and material data supplied for the gearbox. The equations are discussed in Section 2.3.1

In the MATLAB implementation, these formulae are encapsulated in functions that take as input the module, number of teeth, helix angle, face width and material properties for each gear. The script returns the main mesh frequency and converts it to equivalent shaft speeds by relating f_{mesh} to rotational speed via the tooth count:

$$n_i = \frac{60f_{\text{mesh}}}{z_i}, \quad i = 1, 2, \quad (3.1)$$

where z_i is the number of teeth. These theoretical resonance speeds for each stage are later compared to the finite element results and the plant vibration data.

3.1.2 3D Modelling Drive Systems in Solidworks

A detailed three-dimensional model of the twin-drive system and its interface with the rotary kiln was created in SolidWorks from OEM engineering drawings. The model includes the three-stage gearbox, internal shafts, torsion shafts, pinions, girth gear, spring plates, kiln shell section and the main support and thrust rollers. This geometric representation forms the basis of the finite element model used in Abaqus.

Gear Box

The gearbox contains three parallel-shaft stages. The key geometrical data for each gear pair are summarised in Table 3.1. These values are used consistently in both the MATLAB mesh-stiffness calculations and the FE analysis.

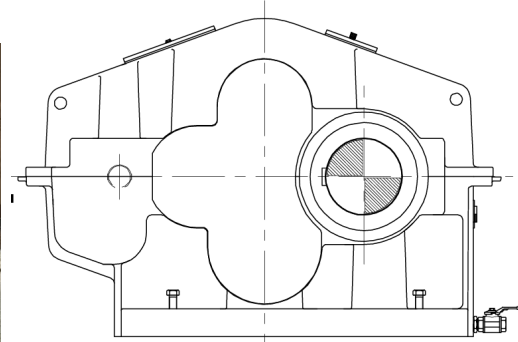
The physical gearbox is shown in Figure 3.1, together with the corresponding CAD model used for meshing.

Number of Teeth		Width (mm)	Module (mm)	Helix Angle (°)	Centre Distance (mm)
Gear 1	Gear 2				
21	61	200	10	15	424.463
20	58	200	12	25	516.381
27	166	400	14	10	646.827

Table 3.1: Gearbox information for the three reduction stages.



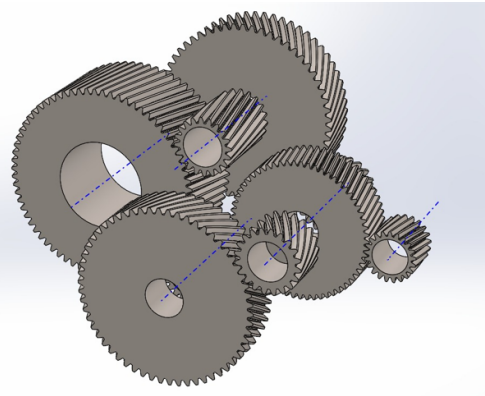
(a) Gearbox housing top half



(b) Gearbox housing schematic



(c) Gearbox isometric view – bottom half



(d) Gearbox CAD

Figure 3.1: Photographs and CAD representation of the gearbox assembly.

Girth Gear

The kiln shell is driven by a large girth gear, which is meshed by a pinion on each side of the twin-drive system. During operation, the kiln shell expands as it heats up; rigidly securing the girth gear directly to the shell would therefore generate large compressive stresses and risk cracking either the shell or the gear web. To avoid this, the girth gear is connected to the shell by a series of spring plates, which are arranged tangentially around the circumference. These plates transmit torque while allowing relative radial expansion between the hot shell and the cooler gear.

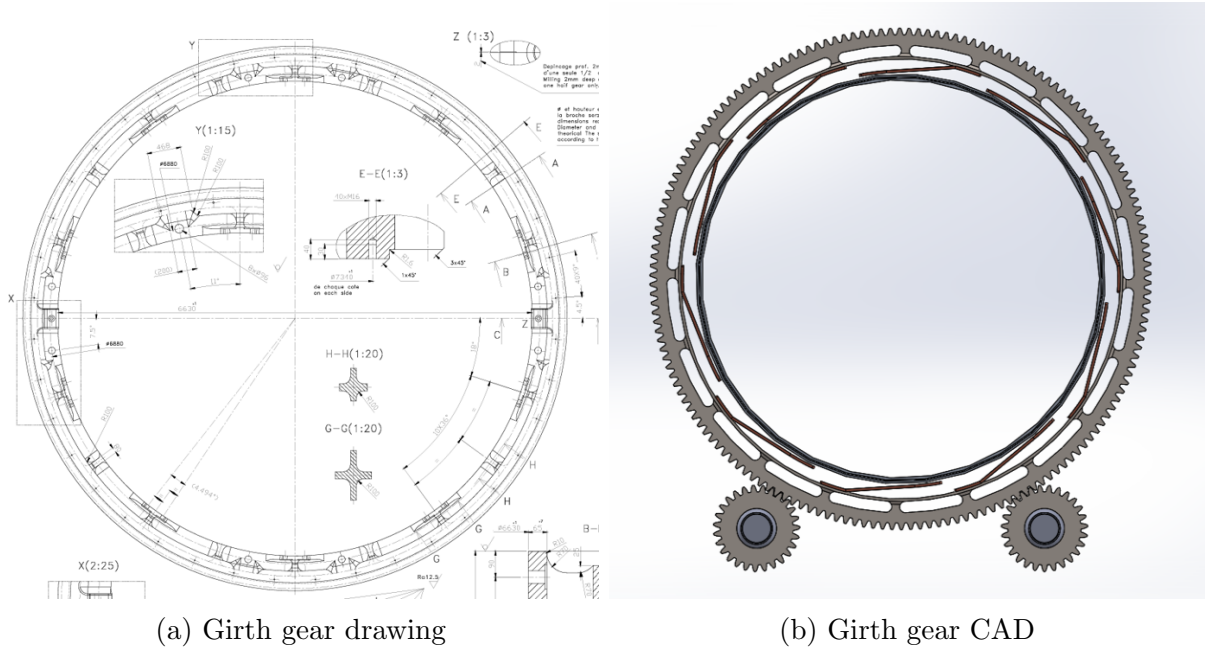


Figure 3.2: Girth gear and its CAD representation.

Spring Plate

The spring plates form the flexible interface between the kiln shell and the girth gear. They are bolted to the shell on their inner ends and to the gear on their outer ends, providing torsional stiffness while accommodating differential thermal expansion.

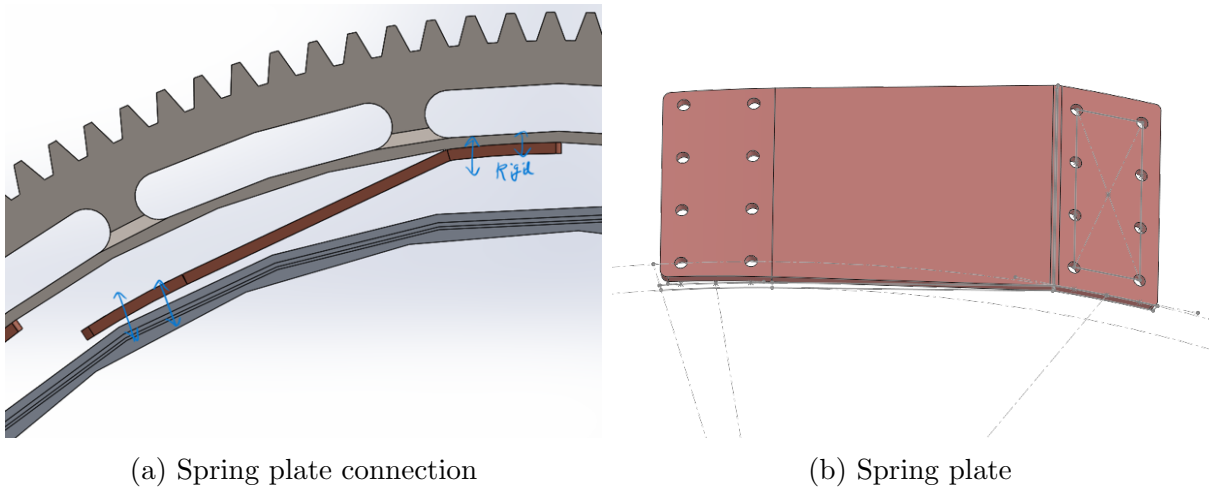
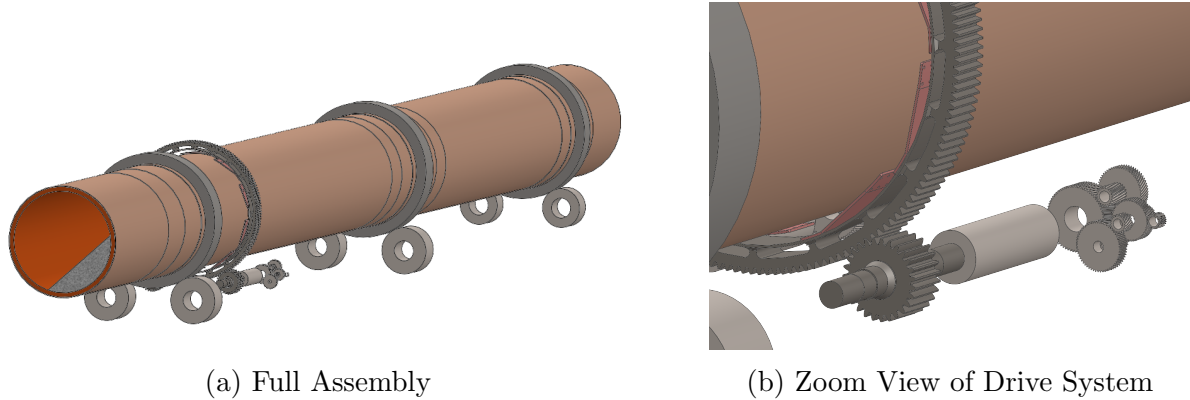


Figure 3.3: Detail of spring plate connection between shell and girth gear.

Kiln Shell & Refractory Assembly

The kiln shell region beneath the girth gear is modelled together with its refractory lining so that the local stiffness and mass distribution are accurately represented in the FE model. The lining consists of multiple grades of bricks (AL-45, AL-60, AL-70 and AL-85) with different densities and dimensions. The full brick schedule is shown in

Table 3.2, and is used to compute the total refractory mass and an effective bulk density for this section of the kiln.



3.2 Simulations

3.2.1 Material Properties

The refractory brick dataset used for the kiln shell region is summarised below. Each entry specifies the brick grade, dimensions and quantity, from which total mass and volume are obtained. Summing over all entries yields a total mass of approximately 1.02×10^6 kg and a total volume of 570.0 m^3 , corresponding to a final bulk density of 2570 kg/m^3 for the lining.

SR	Quality	H	E	C	D	Req. Qty	Total Wt (kg)	Volume (m^3)
1	AL-45(S)	250	198	95.5	86.375	22506	252111.83	109.6138395
2	AL-60(S)	250	198	95.5	86.375	12276	152462.89	59.7893670
3	AL-70(S)	250	198	95.5	86.375	44826	589468.80	218.3217795
4	AL-85	225	198	95.5	87.275	2226	29389.18	10.4961344
Totals						81834	1,023,432.7	398.2211204
Final density:								2570 kg/m^3

Table 3.2: Refractory brick dataset with dimensions and final bulk density.

Component	Material	Density (kg/m^3)	E (GPa)	ν
Gearbox	19CrNiMo6	7800	210	0.30
Kiln shell	P265 GH	7850	164	0.29
Spring plate	P265 GH	7850	164	0.29
Shafts	42CrMo4	7850	200	0.30
Support rollers etc.	CS700	7800	200	0.30

Table 3.3: Material properties used in the model

In the FE model, this effective bulk density is assigned to the refractory volume in the shell segment under the girth gear to reproduce the correct mass and inertia of the lined kiln section.

3.2.2 Boundary Conditions

Gearbox shafts and supports. In the Abaqus model, the gearbox shafts are represented by point masses located at the shaft centres and connected to the gear bores using kinematic couplings. The shafts are assumed rigid in torsion over the frequency range of interest, and their mass and rotary inertia are applied directly at the reference points.

For a shaft of diameter d and length L , made of steel with density ρ , the mass and polar moment of inertia are

$$m_{\text{shaft}} = \rho \pi \left(\frac{d}{2}\right)^2 L, \quad I_{\text{shaft}} = \frac{1}{2} m_{\text{shaft}} \left(\frac{d}{2}\right)^2. \quad (3.2)$$

These are calculated in MATLAB and transferred into Abaqus as point-mass properties.

The gearbox housing is constrained at its foundation pads with elastic supports that approximate the stiffness of the real concrete foundation and anchor bolts. This allows the modal analysis to capture the influence of foundation flexibility, which is consistent with the foundation cracking observed at the plant.

Loaded condition and equivalent kiln inertia. Under loaded conditions, the gearbox output shaft drives the torsion shaft and pinion, which in turn rotate the girth gear and kiln. Rather than modelling every internal component of the kiln in detail, the effect of the rotating shell and granular bed is represented by an equivalent inertia attached to the drive.

The kiln has an internal diameter $D = 5.5$ m and rotates at approximately $n = 3$ rpm with a fill degree $f_{\text{fill}} = 0.18$ and a dynamic angle of repose $\varphi = 32^\circ$. Taking a representative particle diameter $d = 30$ mm, the Froude number is

$$f = 0.18, \quad R = 2.75, \quad H_{\text{bed}} = 1.29 \quad (3.3)$$

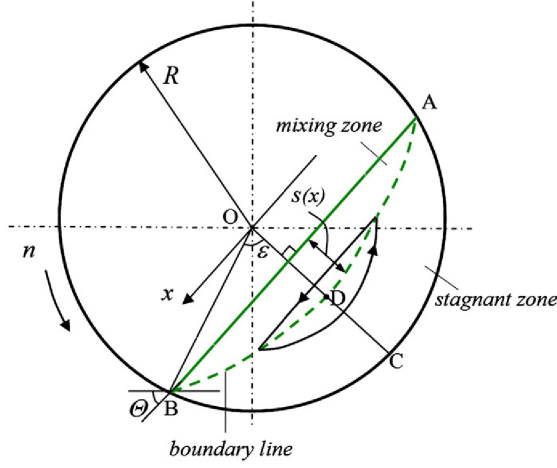
$$\text{Fr} = \left(\frac{2\pi n}{60}\right)^2 \frac{D/2}{g} = 0.027667 \quad (3.4)$$

Using the rolling-bed model of Liu and Specht, the lower boundary of the active mixing layer is defined by the coordinate x_0 (Yan Liu and Specht 2010):

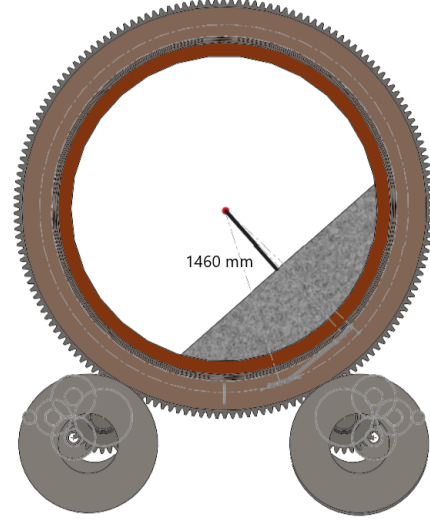
$$x_0 = -\sqrt{\sin^2 \varepsilon - 2(1 + \cos \varepsilon) \frac{d}{D}} = -0.510543 \quad (3.5)$$

where ε is the fill angle. A dimensionless constant k is then computed as

$$k = \frac{\tan \varphi - \tan(1.32 \varphi + 0.32 \varphi f_{\text{fill}} + 10\pi \sqrt{d/D} \text{Fr})}{\sin \varepsilon} = -0.899153 \quad (3.6)$$



(a) Diagram(Yan Liu and Specht 2010)



(b) CAD Construction

Figure 3.5: Rolling Motion of Granular Material Flow

The fraction of the mixing zone (FMZ) in the bed cross-section is obtained by integrating the local thickness of the flowing layer between the limits of the bed:

$$\text{FMZ} = \frac{2}{\pi f_{\text{fill}}} \int_{-\sin \varepsilon}^0 \left[\frac{x - x_0}{\tan \varphi} + \frac{\ln \left(\frac{2 + \sin(2\varphi) k x_0}{2 + \sin(2\varphi) k x} \right)}{k \sin^2 \varphi} + \frac{d}{D} \right] dx = 0.098006 \quad (3.7)$$

Assuming a kiln radius $R = D/2$, a shell length of $L \approx 88$ m, a bulk density of the granular material $\rho_{\text{mat}} = 1800$ kg/m³ and a vertical distance from the kiln centre to the centre of mass of the active layer of $h = 1.46$ m, the effective polar moment of inertia of the rotating bed and shell segment is approximated as

$$I_{zz,\text{eff}} = \pi R^2 L f_{\text{fill}} \rho_{\text{mat}} \text{FMZ} h^2 = 3743.78 \text{ kg m}^2 \quad (3.8)$$

Since there is a gear reduction step of 27/166 at Pinion/Girth Gear interface, the effective polar moment of inertia of the rotating bed and shell segment experienced at output to the gearbox is (Budimir 2019)

$$I_{zz,\text{eff}} = \pi R^2 L f_{\text{fill}} \rho_{\text{mat}} \text{FMZ} h^2 \left(\frac{27}{166} \right)^2 = 141513.86 \text{ kg m}^2 \quad (3.9)$$

These equivalent inertias are then applied in Abaqus as point-mass properties at the torsion-shaft reference point and coupled to the pinion and girth gear through kinematic constraints as shown in figure 3.6. An assumption is made of even load distribution across both drive systems for simplicity. This ensures that the eigenvalue problem solved in the modal analysis reflects the dynamic interaction between the gearbox, torsion shaft, pinion, girth gear, kiln shell and internal granular bed. Contacts have defined allowing dynamic interaction between the gear teeth.

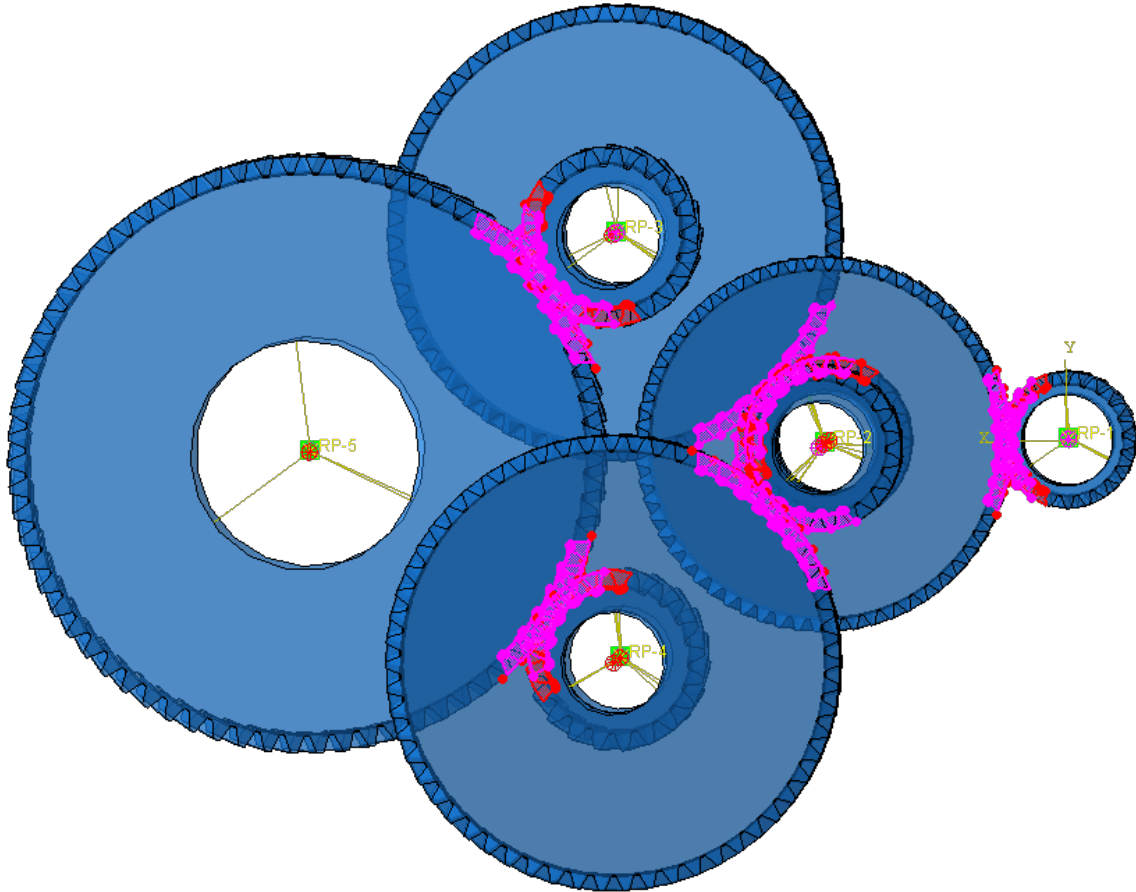


Figure 3.6: Boundary Conditions Set Up

4. Results

4.1 Gearbox

This section presents the finite element modal analysis results for the gearbox, starting from mesh-convergence verification and progressing from individual gear pairs to the complete gearbox in both unloaded and loaded configurations. The results are interpreted in terms of equivalent kiln rotational speeds to assess proximity to the problematic vibration band identified on site.

4.1.1 Individual Unloaded Pairs

The analytical gear-mesh frequency model developed in Chapter 3 was first validated against finite element results for three individual gear pairs corresponding to the three stages of the gearbox. In these simulations, only the gears and their inherent inertias were included; no external components (shafts, housing or kiln) were attached, representing an “unloaded” gearbox configuration.

Grid Independence Study

To minimise numerical uncertainty, a global mesh refinement study was carried out on each gearbox model. The global element seed size was progressively reduced, and the first natural frequency (Mode 1) was monitored at each refinement step.

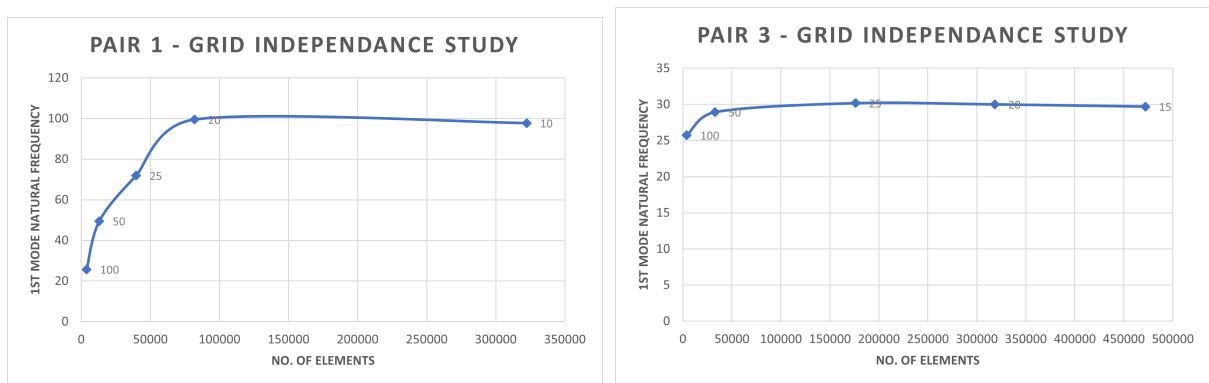


Figure 4.1: Mesh Convergence

The mesh was considered converged when the change in Mode 1 frequency between two consecutive mesh densities fell below 5%. The final refinement produced a variation of less than 2%, indicating that further mesh refinement would not significantly alter

the predicted frequencies. A global seed size of 10 mm was therefore adopted for all subsequent analyses as a good compromise between accuracy and computational cost.

Pair 1 - 61/21

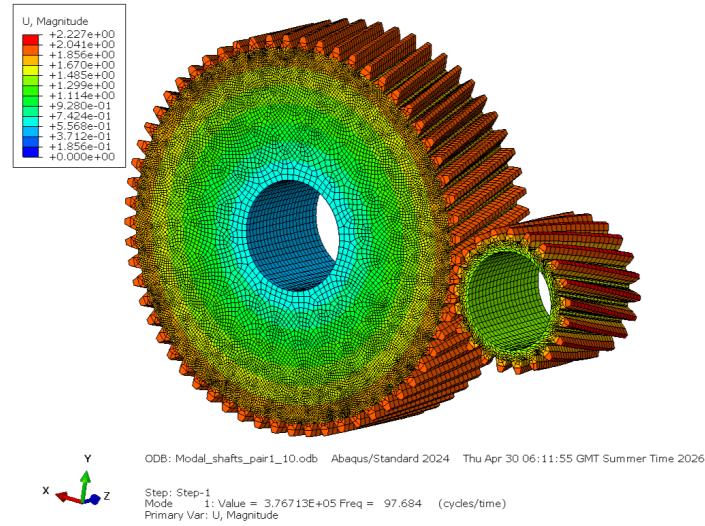


Figure 4.2: Stage 1 - First Mode (Torsional)

$$RPM_{Kiln} = \underbrace{97.68 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{21}{61}}_{\text{Stage 1}} \times \underbrace{\frac{20}{58}}_{\text{Stage 2}} \times \underbrace{\frac{20}{71}}_{\text{Stage 3}} \times \underbrace{\frac{27}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 5.33 \text{ RPM}$$

Pair 2 - 58/20

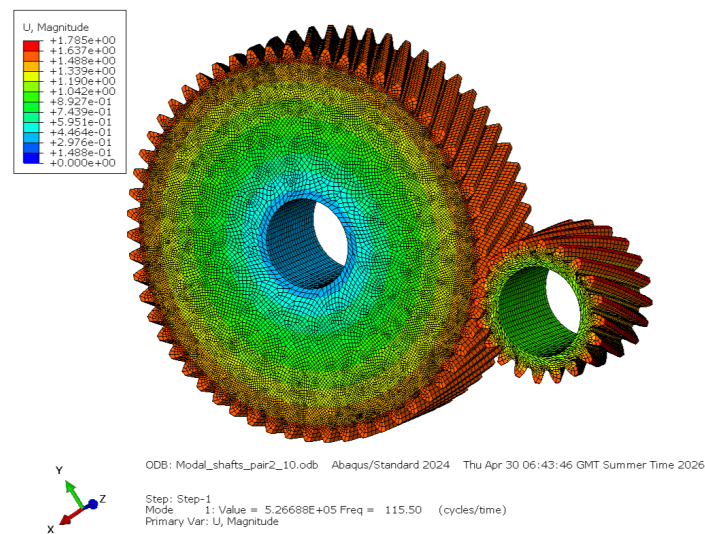


Figure 4.3: Stage 2 - First Mode (Torsional)

$$RPM_{Kiln} = \underbrace{115.50 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{20}{58}}_{\text{Stage 2}} \times \underbrace{\frac{20}{71}}_{\text{Stage 3}} \times \underbrace{\frac{27}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 18.29 \text{ RPM}$$

Pair 3 - 58/20

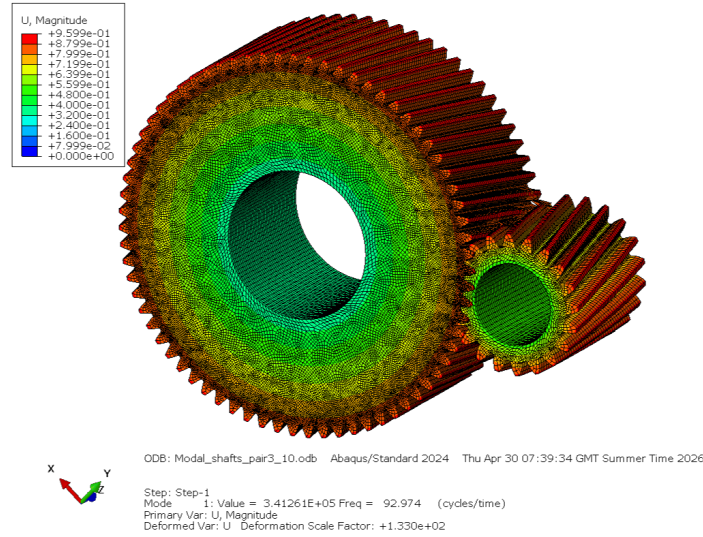


Figure 4.4: Stage 3 - First Mode (Torsional)

$$RPM_{Kiln} = \underbrace{92.97 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{20}{71}}_{\text{Stage 3}} \times \underbrace{\frac{27}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 42.71 \text{ RPM}$$

4.1.2 Summary

For each pair, the first torsional mode was extracted in Abaqus and compared to the ISO 6336-based theoretical prediction. Table ?? summarises the results.

Number of Teeth		Width	Module	Natural Frequency (Hz)		Difference (%)
Gear 1	Gear 2			Theory	FEA	
21	61	0.2	0.010	97.62	97.68	0.06
20	58	0.2	0.012	22.82 <i>2nd harmonic</i> $\times 2 = 45.632$	44.60	2.31
20	71	0.4	0.014	30.29	29.70	1.97

Table 4.1: Comparison of theoretical and FEA natural frequencies for the kiln gear pairs

Across all pairs, the difference between theoretical and FEA torsional frequencies remains within approximately 2.3%, which is acceptable for the present purpose and confirms that the ISO 6336-based stiffness model is capturing the dominant behaviour.

When mapped back through the gear ratios to the kiln, the first gear pair (21/61) corresponds to a kiln speed of roughly 5.07 rpm in the unloaded condition. This lies close to, but slightly above, the observed problematic band of 2.5 rpm to 3.2 rpm. Since adding the rest of the drivetrain and kiln is expected to reduce the natural frequency due to increased inertia, this result already suggests that the first stage is a strong candidate for resonance under loaded conditions.

4.1.3 Full Gearbox

The next step was to assemble the complete gearbox, including all three stages, shafts and housing, and to compute its natural frequencies in the unloaded configuration. The objective was to identify global gearbox modes and to understand how the individual pair modes combine when the system is fully connected.

$$RPM_{Kiln} = \underbrace{82.433 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{21}{61}}_{\text{Stage 1}} \times \underbrace{\frac{20}{58}}_{\text{Stage 2}} \times \underbrace{\frac{20}{71}}_{\text{Stage 3}} \times \underbrace{\frac{27}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 4.50 \text{ RPM}$$

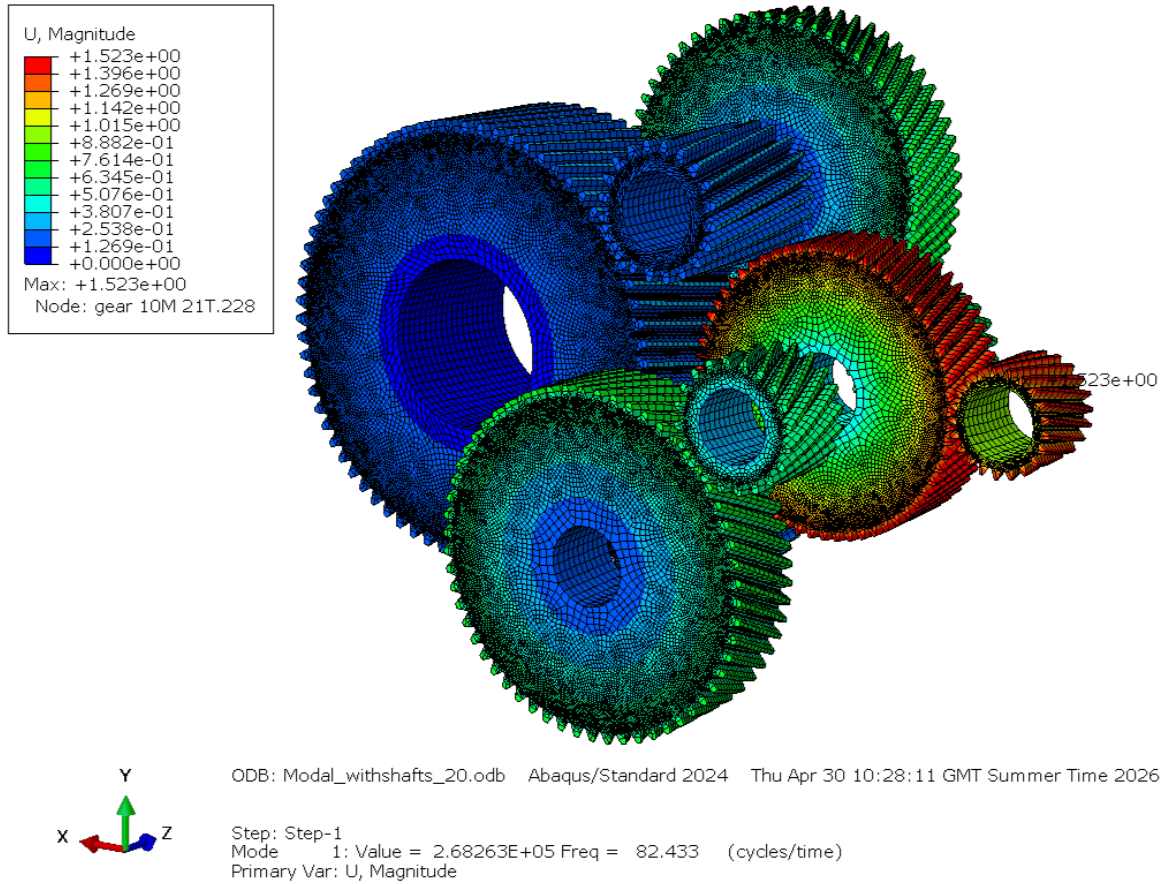


Figure 4.5: Full Gearbox Modal Analysis

The first significant torsional mode of the complete gearbox was found at approxi-

mately 86.3 Hz, which corresponds to a kiln shaft speed of about 4.48 rpm when mapped through all three gear stages. This speed is still outside the critical band observed in plant operation but is relatively close in frequency space. The associated mode shape shows torsional deformation concentrated in the first stage and the input shaft region, with the housing remaining comparatively stiff.

Figure 4.5 illustrates this mode shape, highlighting the regions of maximum angular displacement.

4.1.4 Gearbox – Loaded

To represent real operating conditions, the gearbox model was then coupled to the torsion shaft, pinion, girth gear and the equivalent kiln inertia derived in Section ???. This “loaded” model reflects the additional mass and flexibility introduced by the downstream drivetrain and the rotating kiln with its granular bed.

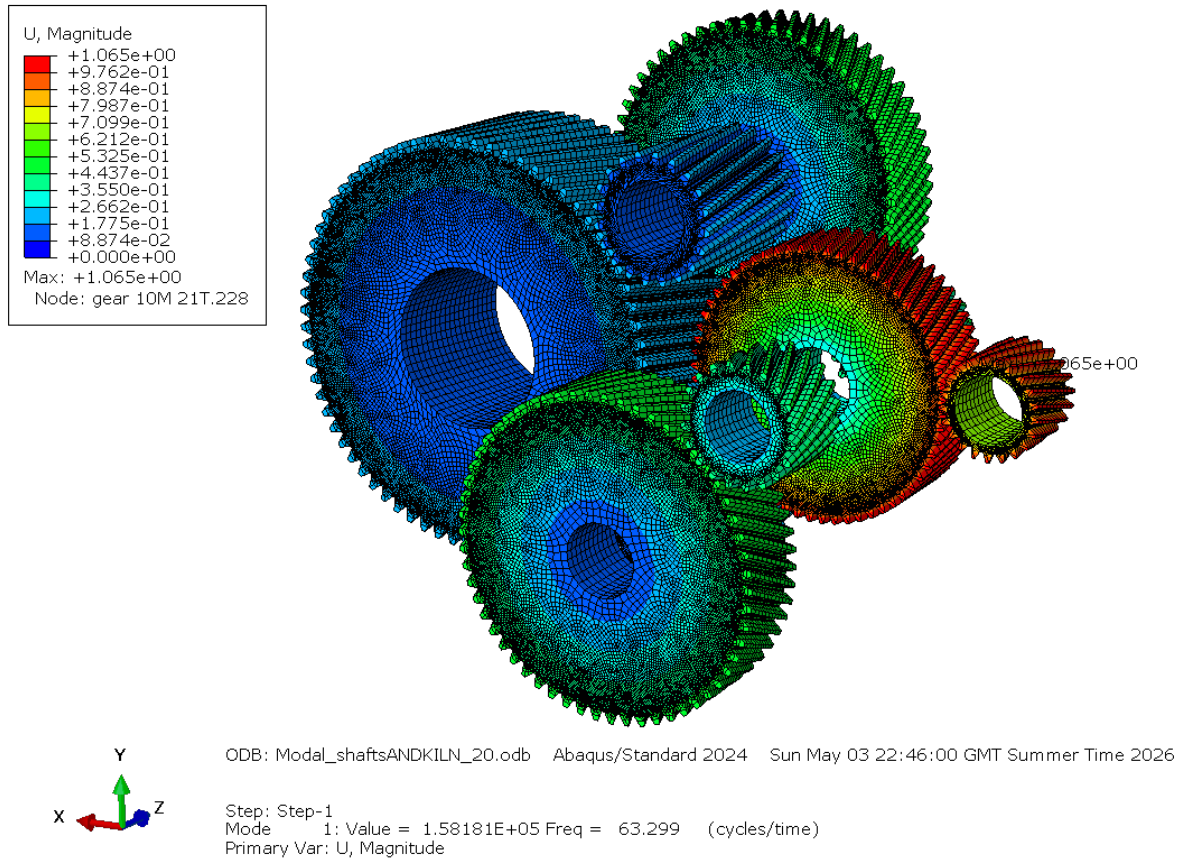


Figure 4.6: Full Gearbox Modal Analysis

$$RPM_{Kiln} = \underbrace{53.486 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{21}{61}}_{\text{Stage 1}} \times \underbrace{\frac{20}{58}}_{\text{Stage 2}} \times \underbrace{\frac{20}{71}}_{\text{Stage 3}} \times \underbrace{\frac{27}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 2.92 \text{ RPM}$$

In this configuration, the dominant torsional mode of interest was found at approximately 45 Hz. When converted to an equivalent kiln speed via the gear ratios, this mode corresponds to around 2.7 rpm, which lies squarely within the problematic band of 2.5 rpm to 3.2 rpm reported on site.

The corresponding mode shape shows strong torsional participation of the first gear-box stage, torsion shaft and pinion, with significant angular deflection at the girth gear interface. This confirms that the first stage gear pair, when coupled to the kiln, resonates within the operating range, providing a plausible explanation for the high vibration amplitudes observed during speed ramps through this band.

Figure 4.6 shows the loaded gearbox mode, with the deformation exaggerated for clarity.

4.2 Torsion Shaft

Finally, the torsion shaft was analysed separately to confirm whether it contributes any critical modes within the operating speed range. The first torsional mode of the shaft–girth gear system was found at approximately 627.62 Hz, which corresponds to a kiln speed of roughly 974.8 rpm. This speed is far above any realistic operating condition for the kiln drive.

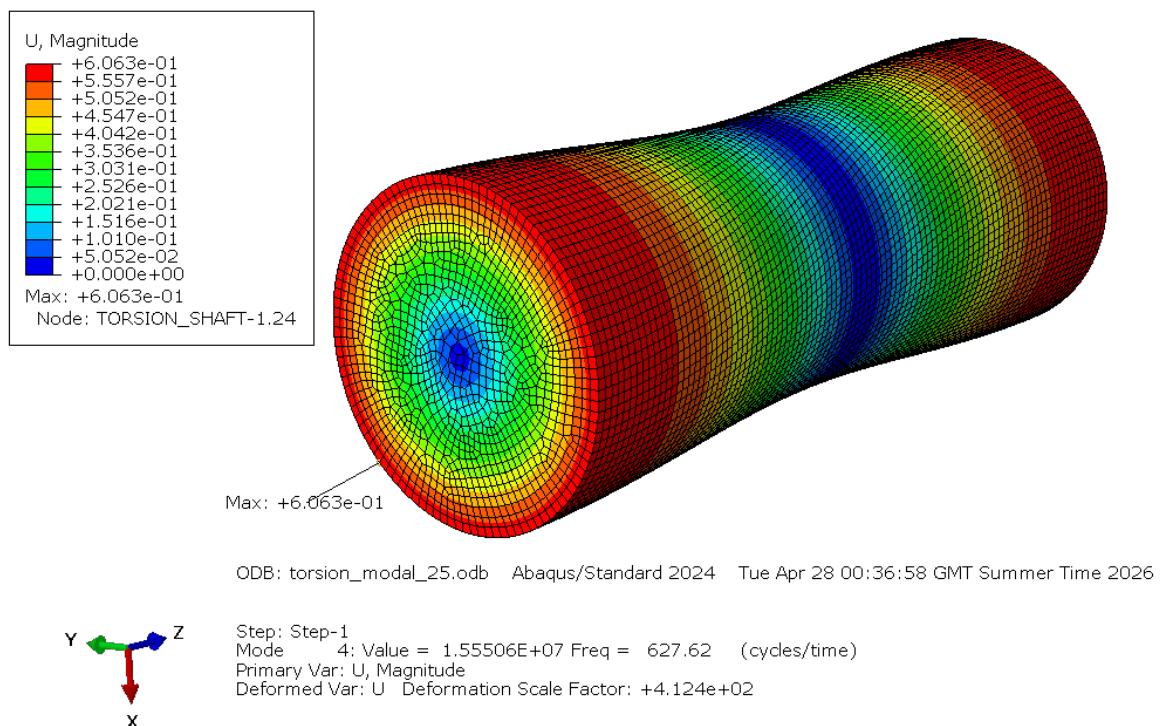
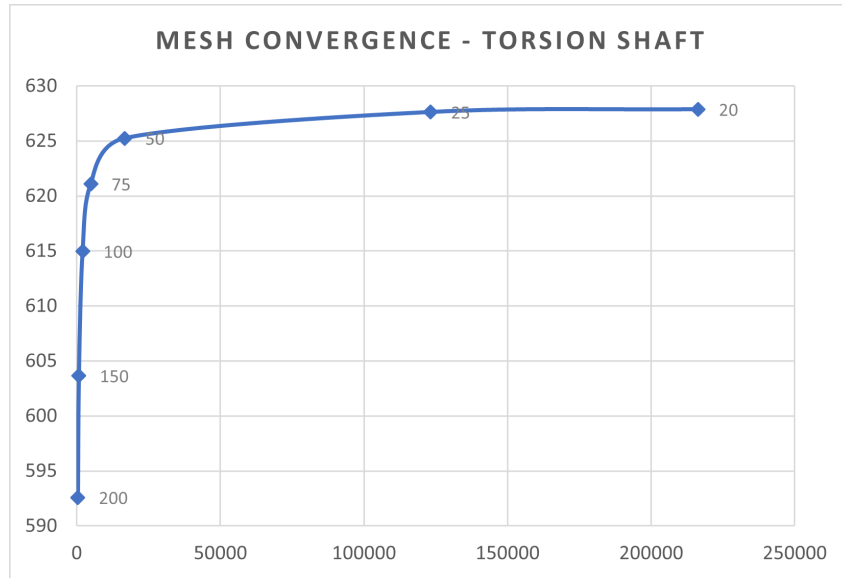


Figure 4.7: Torsion shaft

Figure 4.7 illustrates the torsion shaft mode shape, showing the high-frequency tor-

sional deformation pattern that remains dynamically isolated from the kiln's normal operating speeds.

$$RPM_{Kiln} = \underbrace{627.62 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{27}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 974.8 \text{ RPM}$$



Even when the load of the kiln and granular bed is included, the torsion shaft's natural frequency is not expected to drop anywhere near the 2.5 rpm to 3.2 rpm band. It can therefore be concluded that the torsion shaft is not a critical component from a resonance perspective in the current configuration.

5. Mitigation Strategies

5.1 Introduction

The results in Chapter 4 show that, once the kiln inertia is included, the dominant torsional mode of the drive system aligns with a kiln speed of approximately 2.7 rpm, lying within the problematic 2.5 rpm to 3.2 rpm band observed on site. This mode is dominated by the first gearbox stage and its connection to the torsion shaft and pinion, confirming that resonance is primarily associated with the input side of the gearbox rather than the torsion shaft or downstream components.

Any effective mitigation must therefore:

- shift the critical torsional frequency away from the operating speed range, preferably beyond 5 rpm or below 2 rpm in kiln terms;
- preserve the overall gearbox ratio required by the process (i.e. maintain the input-to-kiln speed relationship);
- remain compatible with the existing gearbox housing, shaft centre distances and foundation geometry, or require only realistic modifications.

This chapter evaluates two mitigation strategies: re-configuring the first gearbox stage using idler gears, and modifying the girth gear pinion tooth count. Each is assessed independently before a combined feasible solution is proposed.

5.2 Modification of Gear Train

5.2.1 Concept and constraints

From the modal analysis it is clear that the first stage (21/61) plays a key role in exciting the resonant mode. One intuitive mitigation is to redesign this stage so that the resonance is moved away from the critical band while keeping the overall gearbox ratio unchanged.

The design objective is therefore to replace the original 21/61 interface by an equivalent two-stage arrangement whose combined ratio remains approximately 61/21, so that the final kiln speed at rated motor speed is preserved. At the same time, the new stages should modify the mass and stiffness distribution so that the first torsional mode shifts out of the 2.5 rpm to 3.2 rpm window.

However, several constraints limit the available design space:

- The centre-to-centre distance between the first and second shafts is fixed at 425 mm by the existing bearing locations in the housing.
- The housing has only limited provision for additional shafts. A dummy box is present below the input, allowing one extra shaft in that region, but there is no provision to introduce shafts from the side or significantly enlarge the casing. This dummy box exists because the 3-stage gearbox (SDNL) and the 4-stage gearbox (SDVL) share the same housing and differ only in internal assembly.

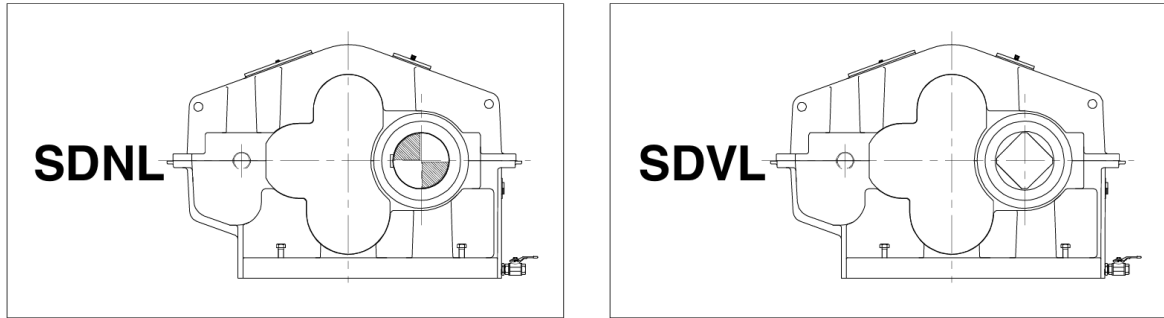


Figure 5.1: Housing Schematics

- Gears must not geometrically interfere; minimum tooth numbers and practical module choices constrain how small or large gears can be.

The possible tooth combinations that satisfy the fixed centre distance of 425 mm and preserve the overall gear ratio are shown in Table 5.1. A MATLAB analytical model was used to predict the resonance speed for each configuration, and selected candidates were validated in Abaqus.

T1	T2	T3	GR	CTC (mm)
25	10	72	0.3472	424.463
24	12	70	0.3429	424.463
23	15	67	0.3433	424.463
22	18	64	0.3438	424.463
21	21	61	0.3443	424.463
20	24	58	0.3448	424.463
19	27	55	0.3455	424.463
18	30	52	0.3462	424.463
17	33	49	0.3469	424.463
15	38	44	0.3409	424.463
14	41	41	0.3415	424.463
13	44	38	0.3421	424.463
12	47	35	0.3429	424.463
11	50	32	0.3438	424.463
10	53	29	0.3448	424.463

Table 5.1: Idler configs satisfying the 425 mm CTC constraint.

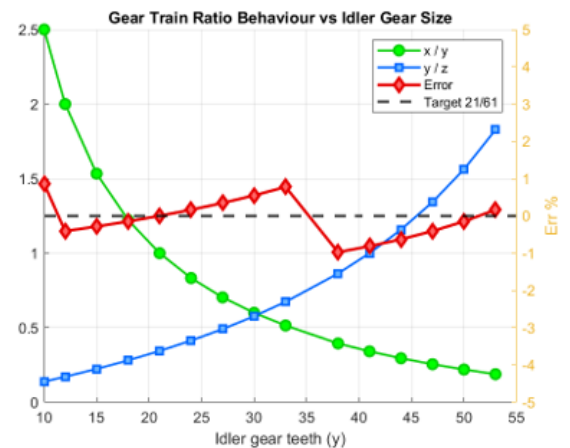


Figure 5.2: Possible Set of Teeth

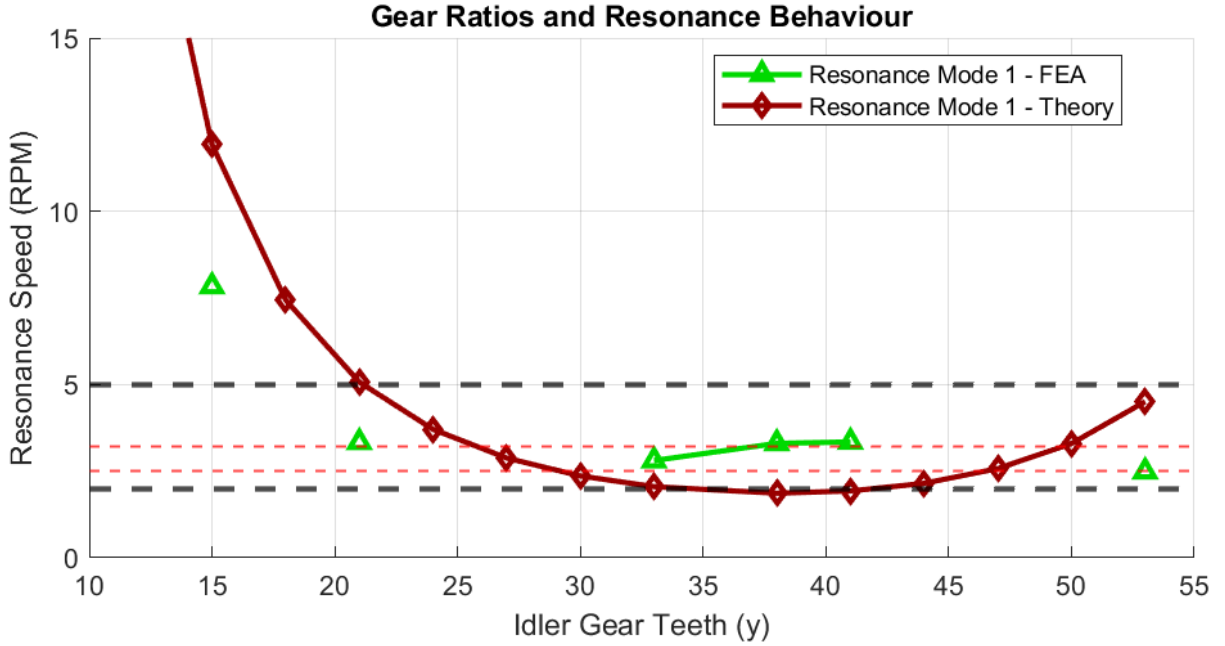


Figure 5.3: Second subfigure

The theoretical sweep shows that introducing a small idler gear significantly increases the resonance speed. However, as the idler tooth count decreases, the physical size of the idler becomes impractically small, leading to geometric interference. The relationship between idler size and predicted resonance speed is shown in Figure 5.3. A monotonic gear train is one in which the output speed consistently increases (ascending) or decreases (descending) with the input, following a single predictable trend.

5.2.2 Big Idler: Non-Monotonic Gear Train

A concept introduced a 53-tooth idler gear, which produces a speed reduction across the first part of the train followed by a speed increase. This configuration makes more effective use of the available centre distance and avoids the extreme tooth counts of the previous concepts.

Using the analytical model, the modified first-stage mesh frequency was converted into an equivalent kiln speed. The resulting natural frequency corresponds to a kiln speed of approximately 2.49 rpm when validated with a focused Abaqus modal analysis. Another observation is that it Theory overestimates FEA results.

Although this represents a shift compared to the original configuration, the critical speed still lies within the problematic 2.5 rpm to 3.2 rpm band. In other words, the resonance is moved slightly but not eliminated; in practice this would simply relocate the worst vibration conditions rather than remove them. For this reason, the Idler 53 concept is considered insufficient as a standalone mitigation.

5.2.3 Small Idlers: Non Monotonic Gear Train

Next concepts investigated used idler gears with 10, 12 and 15 teeth. These configurations aimed to create an initial speed increase followed by a steep speed reduction, thereby moving the critical torsional mode away from the current operating band.

Although the theoretical ratios appeared promising, detailed geometric checks showed that the resulting idler gears became so small that their pitch circles intersected those of the adjacent gears. In practice, the new gears would occupy the same physical space as the existing input and intermediate gears, leading to interference that cannot be resolved without major changes to the housing. Another observation is that it Theory overestimates FEA results.

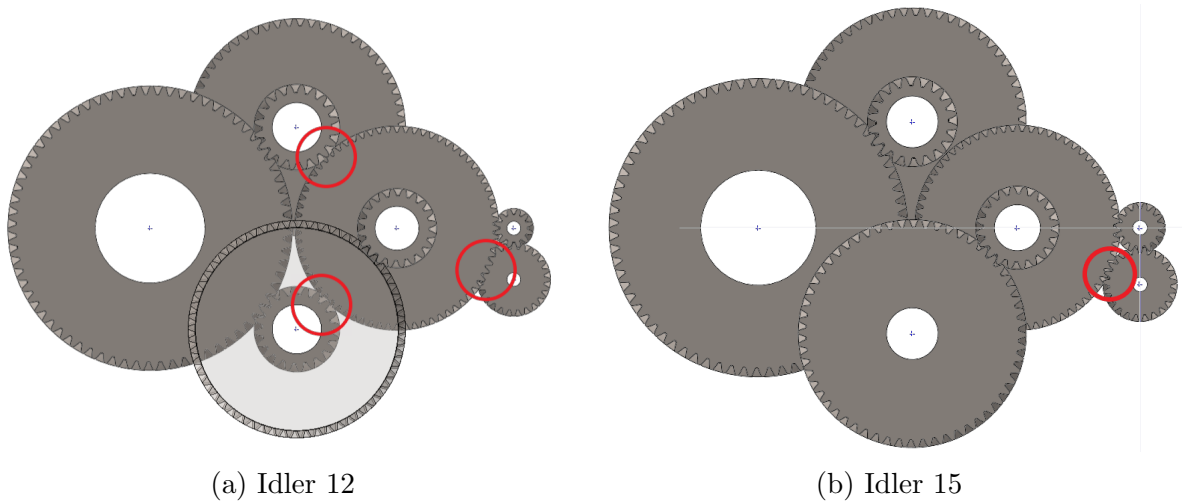


Figure 5.4: Geometric interference observed for small idler configurations.

An alternative orientation in which the idler is driven from the side rather than from directly below the input was also considered. While this improves the gear meshing geometry, the current gearbox housing has no provision for an additional shaft from the side; the only spare space is in the dummy box below the input. Achieving the side-entry solution would require a complete redesign of the housing, which is not considered realistic for a retrofit.

As a result, the 10-, 12- and 15-tooth idler configurations are rejected on geometric and packaging grounds, despite their favourable theoretical effect on natural frequency.

5.2.4 Medium Idler: Monotonic Gear Train

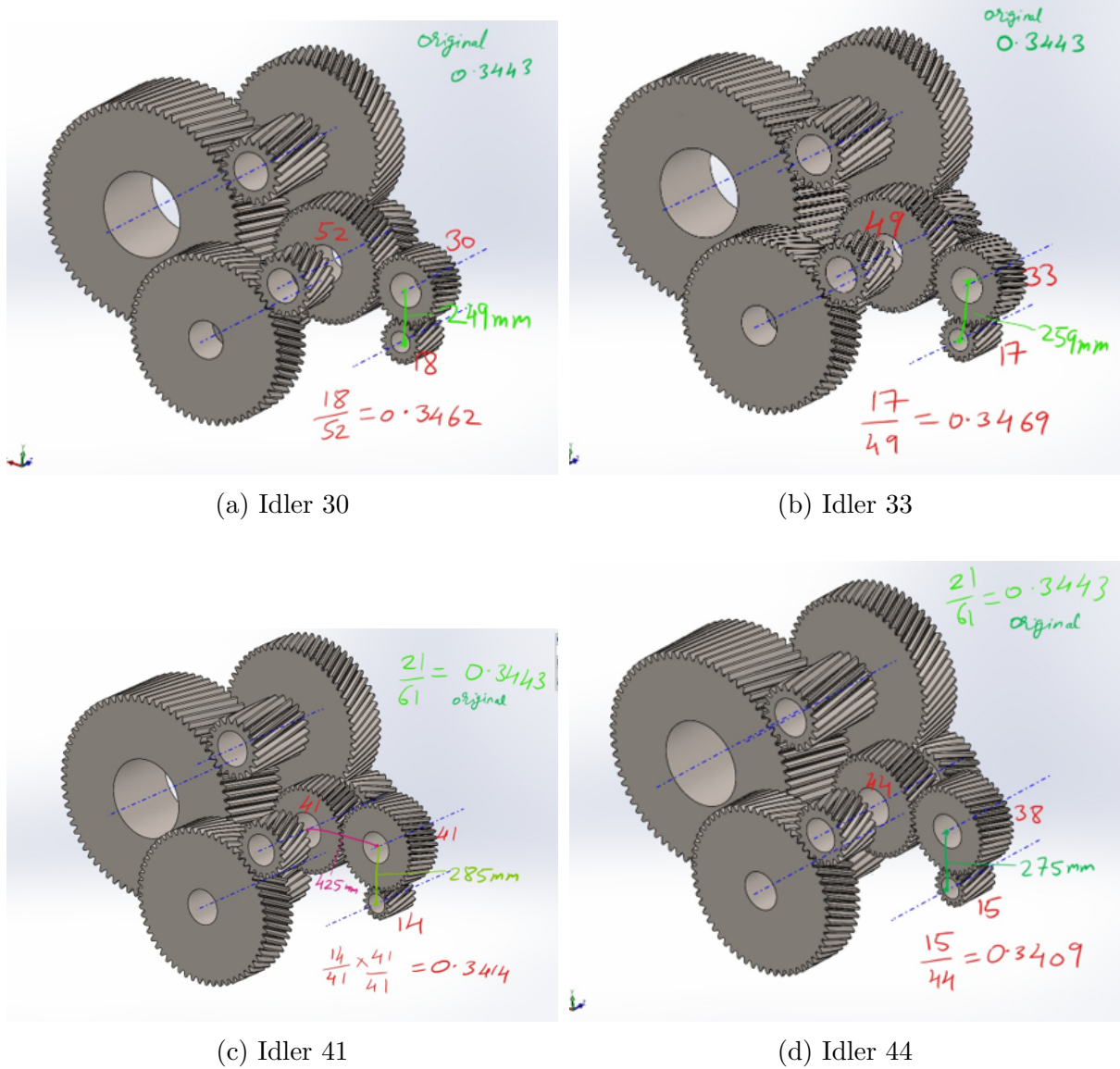


Figure 5.5: Monotonic Gear Train

A third concept used a 41-tooth idler gear ($T_1/T_2/T_3 = 14/41/41$), resulting in a different split of the overall 61/21 ratio between the two new stages. In this case the predicted critical speed shifts upward, with the equivalent kiln speed for the dominant torsional mode becoming approximately 3.34 rpm.

This configuration successfully moves the resonance outside the current 2.5 rpm to 3.2 rpm band. It is also geometrically feasible within the existing housing. However, the shift is marginal — the new critical speed of 3.34 rpm remains within the kiln's normal operating window of 2 rpm to 5 rpm, and transient passages through this speed cannot be avoided entirely during start-up. The idler 41 configuration therefore does not resolve the problem on its own, but it represents the best achievable outcome from gear-train

modification alone and is carried forward as the starting point for the combined solution.

In summary, the idler-based redesigns illustrate that the first-stage resonance can be influenced by re-partitioning the ratio, but the practical geometric and housing constraints prevent achieving a large-enough shift in critical speed within the existing casing.

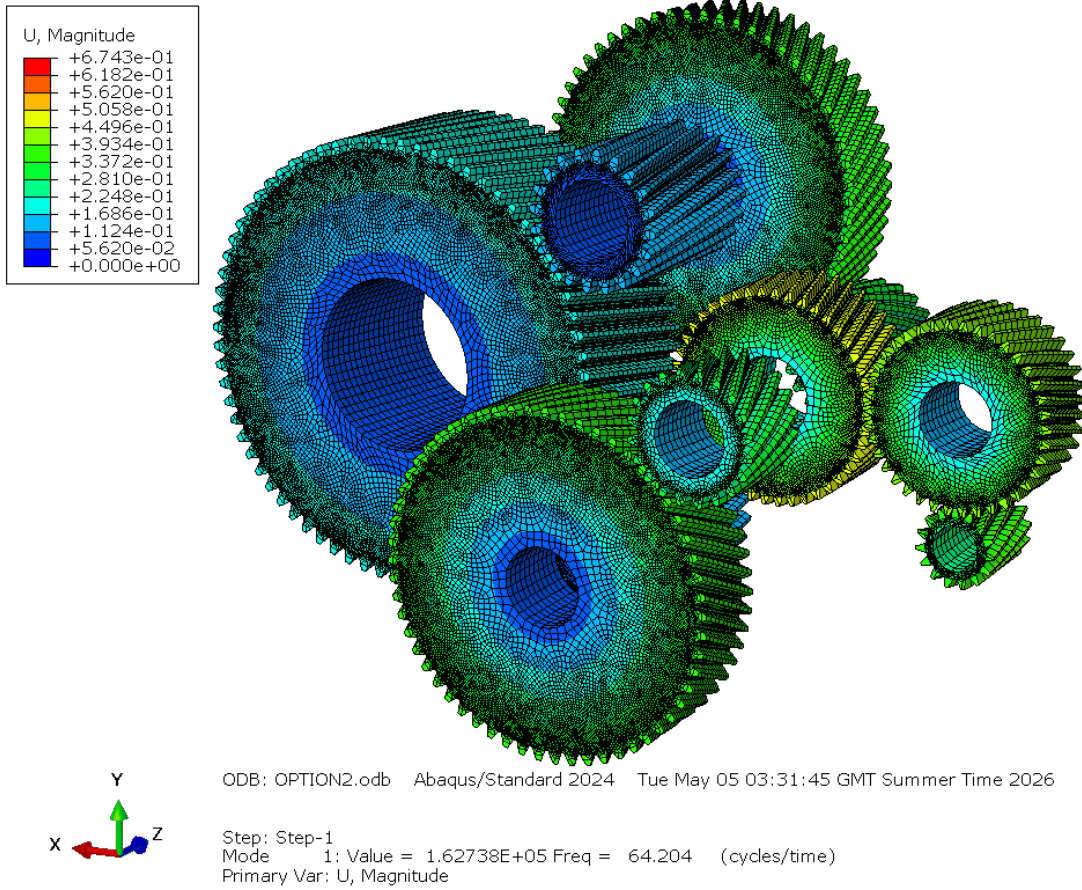


Figure 5.6: Modified Gearbox Modal Analysis

5.3 Modifying Girth Gear Pinion

An alternative approach to shifting the critical speed is to modify the pinion-to-girth gear ratio. Unlike the idler approach, which changes the stiffness and inertia distribution of the gearbox, modifying the pinion tooth count leaves the natural frequency unchanged and instead alters the gear ratio used to map that frequency to an equivalent kiln speed.

From the results in Chapter 4, the current configuration exhibits resonance near 2.8rpm. To shift this to at least 5rpm, the pinion tooth count must be increased such that the same torsional mode frequency corresponds to a higher kiln speed. A simple proportional estimate gives:

$$z_{\text{pinion,new}} \approx \frac{5}{2.9} \times 27 \approx 46 \text{ teeth.}$$

However, increasing the pinion tooth count at fixed module and girth gear tooth count also increases the required centre distance between pinion and girth gear. In the current layout, the centre distance is approximately 4342.5 mm, whereas the new 46-tooth configuration would require approximately 4770 mm — an increase of 427.5 mm.

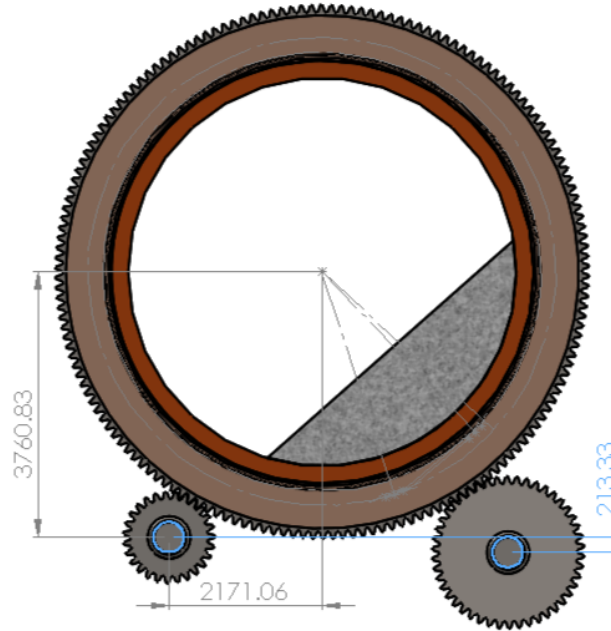


Figure 5.7: Existing(Left) vs Modified Pinion (Right)

The only way to achieve this with the existing girth gear is to lower the pinion housing by removing material from the concrete foundation beneath it. A geometric estimate shows that approximately 213 mm of concrete would need to be removed, representing approximately 37% of the current foundation depth of 574 mm as shown in fig 5.8:

$$\frac{213}{574} = 37.2\%.$$

On-site engineers have confirmed that a maximum of 150 mm is physically available through removal of the foundation cover plate. Since the required removal of 213 mm exceeds this limit:

$$213 \text{ mm} > 150 \text{ mm},$$

the standalone pinion modification is not achievable without major and destructive civil works to the foundation. This raises serious concerns regarding structural stability and alignment accuracy, and is therefore rejected as a standalone solution.

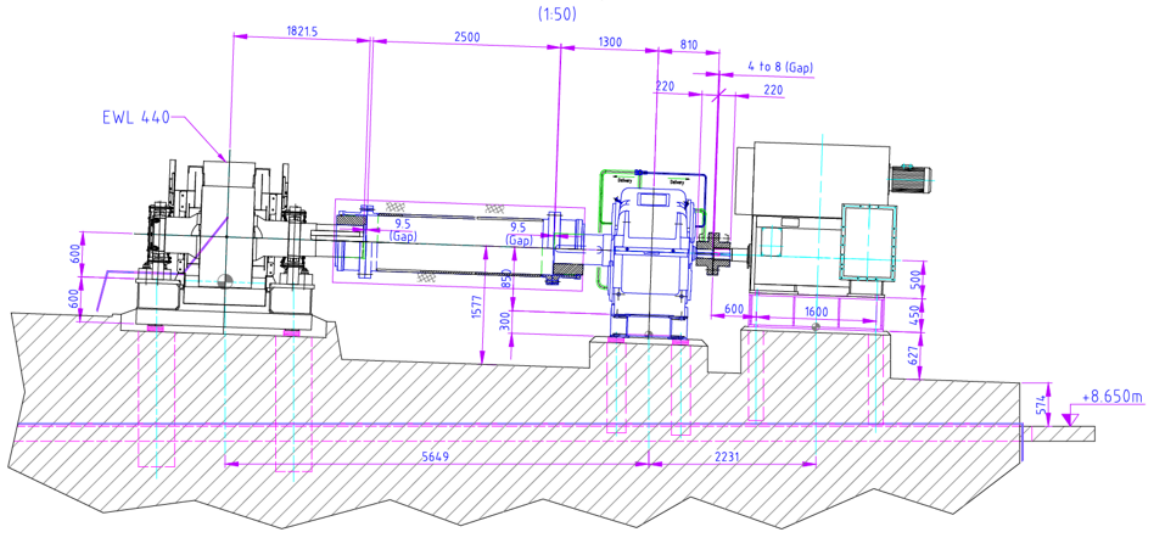


Figure 5.8: Side View of Drive System

5.4 Feasible Solution

Neither strategy alone is sufficient: the idler modification (Idler 41) moves the resonance to 3.34 rpm but does not clear the operating range, and the standalone pinion modification requires 213 mm of foundation removal, which exceeds the 150 mm available on site.

However, combining both strategies produces a viable outcome. Taking the Idler 41 configuration ($T1/T2/T3 = 14/41/41$) as the new gearbox first stage raises the natural frequency from 53.486 Hz to 64.910 Hz. The corresponding resonance speed at the existing pinion of 27 teeth is 3.34 rpm. To push this above 5 rpm, the required new pinion tooth count is now:

$$z_{\text{pinion,new}} \approx \frac{5}{3.34} \times 27 \approx 40 \text{ teeth.}$$

This smaller increase in pinion teeth — from 27 to 40 rather than 27 to 46 — requires a correspondingly smaller increase in centre distance, from approximately 4342.5 mm to approximately 4635 mm, an increase of 292.5 mm. The required foundation removal is now approximately 145 mm. Critically, this satisfies the on-site constraint:

$$145 \text{ mm} < 150 \text{ mm.}$$

The combined solution is therefore geometrically and civilly feasible within the existing installation. The new configuration is validated in Abaqus, giving a dominant torsional mode natural frequency of 64.910 Hz, which maps to a kiln speed of 5.07 rpm — outside the operating window:

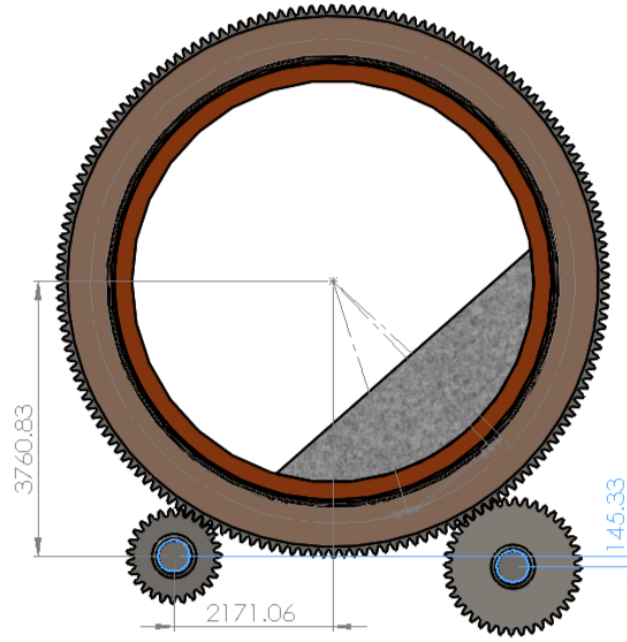


Figure 5.9: Existing(Left) vs Modified Pinion (Right)

$$RPM_{Kiln} = \underbrace{64.910 \text{ Hz}}_{\text{Natural Frequency}} \times \underbrace{\frac{21}{61}}_{\text{Stage 1}} \times \underbrace{\frac{20}{58}}_{\text{Stage 2}} \times \underbrace{\frac{20}{71}}_{\text{Stage 3}} \times \underbrace{\frac{40}{166}}_{\text{Pinion/Girth Gear}} \times \underbrace{\frac{60}{2\pi}}_{\text{Hz} \rightarrow \text{RPM}} = 5.07 \text{ RPM}$$

The mode shape of the new configuration confirms that the dominant torsional mode no longer lies within the 2 rpm to 5 rpm operating window. The detailed design of this combined solution would require collaboration with the gearbox OEM and structural engineers to finalise gear geometry, bearing selection, and foundation modification. The analytical and FE work presented here establishes the quantitative targets: a first-stage configuration of $T1/T2/T3 = 14/41/41$ and a pinion tooth count of 40, requiring approximately 145 mm of foundation lowering.

Component	Existing	Proposed	Implementation
T1	N/A	14	Provision of additional input shaft in housing.
T2	21	41	On the same shaft, replace gear.
T3	61	41	On the same shaft, replace gear.
Pinion	27	40	Repositioning of pinion shaft and housing, and therefore the drive system, by 145 mm.

Figure 5.10: Summary of Design Changes

6. Discussions

6.1 Overview of Key Findings

This project investigated the cause of speed-dependent vibrations in the RC04 twin-drive rotary kiln and explored practical ways to reduce them within the limits of the existing gearbox housing and foundations. The analysis showed that the vibration problem at around 2.5–3.2 rpm is caused by a torsional vibration mode in the first stage of the gearbox. The torsion shaft itself was found not to create critical vibrations within the operating range.

The simplified ISO 6336 gear-mesh stiffness model matched the Abaqus finite element results within about 2.3%, showing that the reduced model accurately represents the gearbox dynamics. Leveraging this validated model, design decisions were informed and optimized through targeted simulations, eliminating the need for exhaustive brute-force analyses and significantly reducing overall computational effort and project time.

An important part of the work was modelling the kiln and the material inside it as an equivalent rotational inertia, based on granular flow behaviour rather than simple static calculations. Using this approach, the main torsional mode corresponded to a kiln speed of about 2.7 rpm, which agrees well with measured plant data and earlier Vibrotech studies.

Several mitigation options were then studied, including alternative idler gear arrangements and changes to the pinion gear size. The final recommended design uses a first-stage idler arrangement of $T1/T2/T3 = 14/41/41$ together with a 40-tooth pinion. This moves the critical speed to approximately 5.07 rpm while still fitting within the current gearbox and foundation constraints.

6.2 Position Within Engineering Practice

This project demonstrates how vibration theory, gear design standards, and finite element analysis can be combined to solve real industrial drivetrain problems. Rotary kilns in cement plants operate continuously, so resonance within the operating range can seriously reduce component life, increase downtime, and waste energy. Instead of simply avoiding certain operating speeds, this work identifies the exact vibration mechanism causing the problem. This allows targeted design improvements rather than repeated replacement of damaged parts.

The modelling approach also reflects modern engineering practice. A simplified analytical model was first developed using ISO 6336 gear stiffness equations, and then vali-

dated using detailed 3D finite element analysis. This combination provides both physical understanding and accurate prediction of gearbox behaviour.

The project also reflects the increasing use of virtual prototyping and digital-twin methods in industry, where engineers test modifications digitally before carrying out expensive plant shutdowns or structural changes.

6.3 Relationship to Existing Research

The findings of this study are consistent with the broader gearbox dynamics literature, which shows that relatively small changes in stiffness, inertia, and support conditions can produce measurable shifts in torsional and bending natural frequencies. (Han et al. 2017; Chen et al. 2016; X. Yang et al. 2023) This agreement is important because it confirms that the observed resonant behaviour is not an isolated result, but part of the expected response of a coupled gear–shaft–bearing system when realistic operating parameters are introduced.

A key contribution of the present work is the treatment of the driven equipment as a physically meaningful dynamic subsystem rather than a simplified rigid inertia. Many gearbox studies idealise the load side as a lumped rigid mass, but the present results show that the kiln, its drive train, and the granular material inside it alter the effective inertia and therefore the predicted critical frequencies (Anonymous 2020; Marafona et al. 2021). The boundary-condition-based methodology developed here is therefore validated not only by the modal results, but also by its ability to reproduce the change in resonance trend observed in the real system.

The present study also strengthens confidence in standards-based stiffness modelling. ISO 6336 methods have previously been shown to match experimental gearbox stiffness and vibration behaviour with errors of up to around 6% in representative cases, making them useful for preliminary design and first-pass dynamic prediction. (Sainsot and Vexlex 2020; Otto, Fromberger, and Stahl 2024) However, many of those studies relied primarily on analytical or experimental comparisons, with less emphasis on finite element validation of the full drivetrain response. By combining ISO 6336-based stiffness estimation with finite element analysis, this work closes that gap: the theoretical model agrees with the finite element results to within 2.3%, while the experimental agreement confirms that the same model captures the measured system behaviour with good accuracy. In this sense, the finite element model is expected to be slightly closer to the full system response because it incorporates realistic geometry, support flexibility, and load-path effects that are not represented in the simplified analytical model.

Finally, the methodology used here is also validated for the granular-material and kiln-loading problem. The finite element boundary conditions developed for the driven equipment reproduce the expected reduction in critical frequency under increased load,

showing that the structural model and the loading representation are consistent with the real operating scenario. This gives confidence that the same workflow can be used not only for gearbox stiffness and resonance prediction, but also for systems in which the driven machine and its contents contribute significantly to the dynamic response.

6.4 Evaluation of Mitigation Strategies

The mitigation study moved from idealised concepts to solutions that were actually practical for the existing installation. Some idler configurations produced large improvements in the simplified model, but many were impossible to implement because of gearbox space limitations, bearing support issues, or the fixed 425 mm centre distance.

A similar issue appeared with increasing the pinion size. A large increase in pinion tooth count could move the critical speed above the operating range, but it would require excessive foundation removal. Increasing the pinion to around 46 teeth would require removing about 213 mm of foundation depth, which exceeds the 150 mm limit accepted by plant engineers.

The final proposed solution combines a moderate idler modification in the Gearbox with a 40-tooth pinion at the Girth Gear. This raises the critical speed to about 5.07 rpm while requiring only about 145 mm of foundation modification, making it both dynamically effective and practically achievable.

6.5 Limitations of the Study

Although the results are consistent, several limitations should be recognised.

First, the study is entirely computational. No experimental modal testing was performed to fully validate the proposed solution. Therefore, factors such as foundation degradation, joint looseness, or temperature effects may still influence the real system. (Safaeifar and Karimi 2015)

Second, the finite element analysis assumes linear elastic behaviour and small vibration amplitudes. Effects such as backlash, non-linear gear contact, or loss of contact under heavy excitation were not included.

The kiln material and refractory lining were also simplified as an equivalent inertia rather than being modelled directly. This means interactions between shell bending, tyre-roller contact, and material movement were not considered.

In addition, the ISO 6336 stiffness model does not include detailed manufacturing effects such as profile modifications or gear micro-geometry. Damping was also not measured directly, so the study predicts critical speeds rather than exact vibration amplitudes.

Because of these limitations, the predicted frequencies should be treated as engineering

estimates rather than exact values. The proposed design should therefore be further validated through detailed OEM design work and on-site testing before implementation.

6.6 Implications and Future Work

The main outcome of this project is that the resonance problem can likely be solved through targeted modification of the drive system rather than continued operational restrictions or repeated component replacement.

The proposed Idler 41 and 40-tooth pinion configuration is predicted to move the dominant torsional mode outside the normal 2–5 rpm operating range. This could reduce the risk of gear and foundation cracking, improve reliability, and reduce costly kiln shutdowns.

The modelling workflow developed in this project could also be applied to other large industrial drivetrains such as ball mills or crushers, where similar problems have been observed at higher speeds of 15 RPM. Future work could improve the model by including measured damping, non-linear gear contact, temperature-dependent material properties, and experimental validation through modal testing. These improvements could help develop broader design guidelines for avoiding resonance in heavy industrial gearbox systems.

7. Impact Assessment

7.1 Introduction

The cement industry plays a vital role in modern infrastructure, supplying materials essential for housing, transport and industrial development. However, maintaining efficient and reliable cement production remains a major engineering challenge due to the scale and continuous nature of plant operations. This project investigates vibration-induced resonance in the twin drive system of a rotary kiln at a cement plant, in India. Using modelling and finite element analysis, the study aims to identify the source of excessive vibration and evaluate methods to improve the mechanical stability of the kiln. Improving the reliability of this critical system can enhance operational efficiency, reduce downtime and support safer long-term plant operation.

7.2 Environmental Impact

The cement industry is responsible for roughly 8% of global CO₂ emissions and is widely recognised as a hard-to-abate sector due to the combination of high-temperature thermal energy demand and unavoidable process emissions from clinker formation (Volaity et al. 2025). The rotary kiln is the single largest energy consumer in the process; clinker burning can account for around 80% of a plant's thermal energy demand, and energy costs are a major share of total production cost. When the twin drive system suffers from resonance at particular speed ranges, operators may be forced to avoid those regimes, running the kiln away from its most thermally efficient operating point and accepting increased wear and risk. More importantly, severe vibration raises the probability of unplanned shutdowns to protect gears, bearings and the refractory lining. Each forced stop is environmentally expensive: the kiln must be cooled, often for many hours, and then reheated slowly to avoid thermal shock, consuming large quantities of fuel with no clinker production. Therefore, this project directly enables sustainable operation allowing business deliveries by reducing operational costs and reducing carbon emissions.

7.3 Societal Impact

Cement production is closely linked to economic development because cement is a fundamental material used in infrastructure, housing and transportation projects. Reliable plant operation therefore has wider societal importance, particularly in developing

regions where large-scale infrastructure growth depends on stable and affordable material supply. Improving the performance and reliability of critical equipment such as rotary kilns can help maintain continuous production, reduce operational costs and support long-term industrial productivity, and therefore better infrastructure for the people.

The safety impact of this project is even more significant. Rotary kilns are extremely large industrial systems operating under harsh thermal and mechanical conditions. Excessive vibration in the drive train increases the likelihood of equipment failure, refractory lining damage and emergency shutdowns. When such failures occur, maintenance workers may need to enter the kiln to inspect or repair internal components. These activities are classified as confined-space work and are considered high-risk due to elevated temperatures, limited ventilation, dust exposure and restricted movement inside the kiln shell.

7.4 Ethical

The main ethical considerations associated with this project involve confidentiality, responsible data handling and professional integrity. The study relies on access to industrial information including vibration data, operational measurements and engineering details related to the kiln drive system. Much of this information is commercially sensitive because it reflects plant performance, equipment configuration and operational practices that are not publicly available.

Therefore, all plant data and technical information are handled responsibly and used only for the purposes of the project. Detailed engineering drawings, plant layouts and sensitive operational values should not be reproduced or shared without permission from the company. Any figures, graphs or schematics included in the final dissertation should be simplified where necessary to avoid disclosing confidential information. Maintaining confidentiality is essential both for protecting the company's intellectual property and for preserving professional trust between the student and the industrial partner.

7.5 Conclusion

Hence, this project investigates a specific mechanical problem within the rotary kiln drive system, but its implications extend far beyond vibration analysis alone. Moreover, the methodology developed here can be implemented to other drive systems, not just a specific case as discussed in this paper.

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A. Appendix A

A.1 Initial Gantt Chart

The initial plan (October 2025), Figure A.1 organised the project based on software, which lacked clarity in the structured approach to the project. The plan was rushed and did not allocate time for essential tasks such as developing skills to use Abaqus and LaTeX. External commitments were not taken into account, and there was no provision for them within the schedule. Furthermore, the initial plan lacked specific deadlines for deliverables, leaving key phases undefined. No time was allocated for a grid independence study before starting the modal analysis, which is crucial for ensuring the validity of simulations.

[illegible]

Figure A.1: Initial Gantt Chart (October 2025)

A.2 Updated Gantt Chart

In the revised Gantt chart, Figure A.2, the project has been streamlined in a "series manner" by dividing into Modelling, Simulation, Validation & Mitigation stages. The updated plan includes time for learning Abaqus and LaTeX, as well as accounting for external commitments that were previously overlooked. The deadlines for all deliverables are now specified, ensuring a clearer roadmap for the project. Time has been allocated for the grid independence study prior to the Modal Analysis simulation step. The case-specific mitigation step has been revised to 30 days, recognising the need for additional time to perform simulations to ensure resonance frequencies are adequately altered out of operational range.

Task	Start Date	End Date	Duration	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##	##
Literature Review & LaTeX				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31				
Understanding Problem and Reading Vibrational Reports	29-Sep	13-Oct	14																																			
Submit Initial Project Plan		10-Oct																																				
Understanding Modal Analysis - FEA Technique	13-Oct	23-Oct	10																																			
Understanding Gear Teeth Stiffness Model	23-Oct	02-Nov	10																																			
Developing writing skills in LaTeX	02-Nov	07-Nov	5																																			
Submit Detailed Project Plan		07-Nov																																				
Modelling																																						
Implementing the Gear Stiffness Model in MATLAB and Solving	07-Nov	19-Nov	12																																			
Analysing Theoretical Gear pair Critical frequencies	19-Nov	24-Nov	5																																			
Creating 3D models of components from Drawings	24-Nov	06-Dec	12																																			
Submit Literature Review		28-Nov																																				
Assembling components and exporting to Abaqus	06-Dec	10-Dec	4																																			
Simulation																																						
Developing advanced simulation skills in Abaqus	10-Dec	17-Dec	7																																			
Solving and Applying Boundary Conditions	17-Dec	27-Dec	10																																			
Grid Independence Study	27-Dec	31-Dec	4																																			
Modal Analysis	31-Dec	23-Jan	23																																			
Frequency Response Study	23-Jan	06-Feb	14																																			
Validation																																						
Analysing Simulation Result to confirm or rule out Resonance	06-Feb	14-Feb	8																																			
Identify Resonating Component	14-Feb	20-Feb	6																																			
Submit Poster		23-Feb																																				
Mitigation Strategies																																						
Suggest case-specific vibration mitigating strategies	20-Feb	07-Mar	15																																			
Re-run the simulation with the suggested changes to confirm	07-Mar	22-Mar	15																																			
Report																																						
1st Draft	22-Mar	01-Apr	10																																			
Submit Sample Writing		27-Mar																																				
2nd Draft + Extensive Marginal Time allowed to account for Delays	01-Apr	24-Apr	23																																			
Proof Read	24-Apr	01-May	7																																			
Submission		01-May																																				

Figure A.2: Revised Gantt Chart (November 2025)

A.3 Gantt Chart – Start of Semester 2

During the first semester, it became clear that ideal planning was challenging due to commitments to other modules and exam preparation. Consequently, the Gantt chart was updated to incorporate additional coursework deadlines, and tasks were reassigned with appropriate time allocations to reflect these constraints.

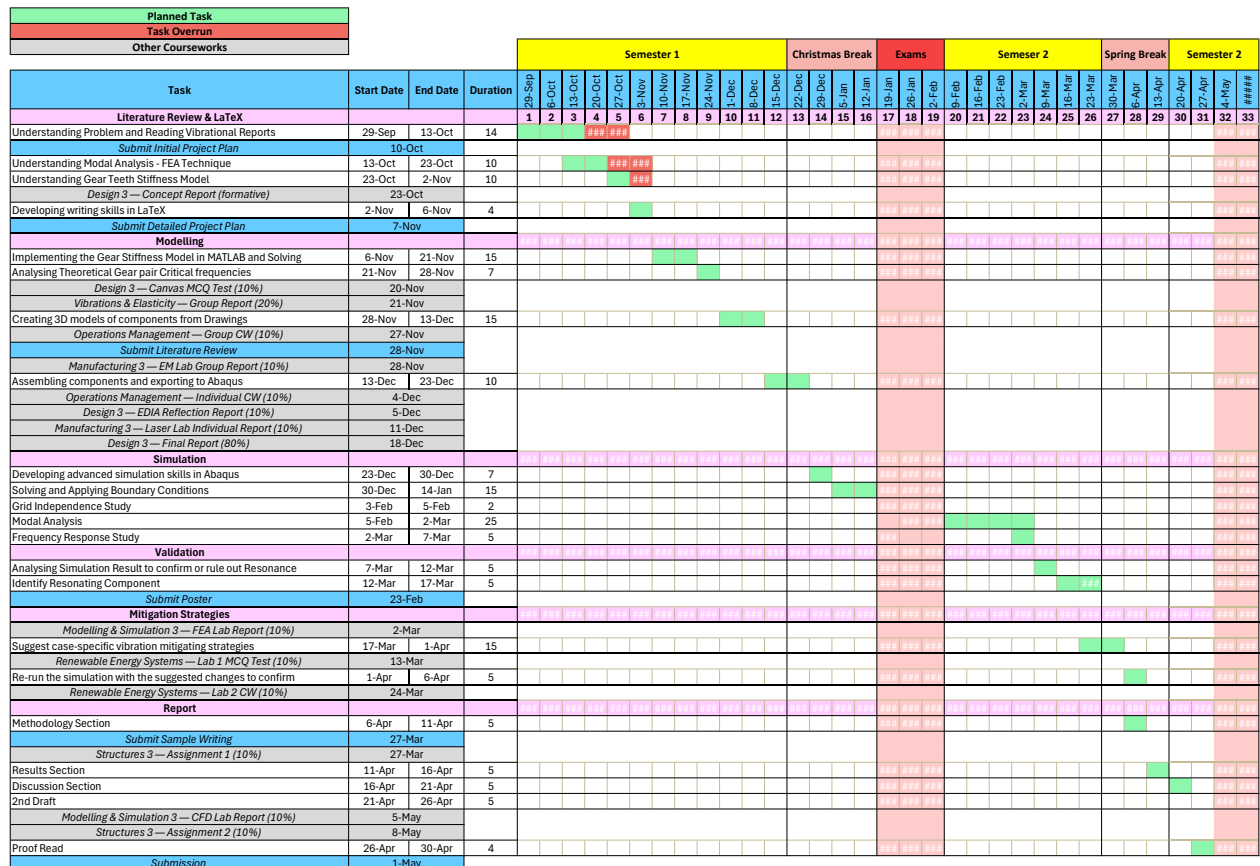


Figure A.3: Start of Semester 2 - Gantt Chart (January 2026)

A.4 Gantt Chart – Final

The final Gantt chart accounts for the double extension, moving the report submission date to 15th May. Tasks that were overrun are indicated by red highlights. These delays were primarily due to unplanned interviews, additional coursework, and periods of illness.

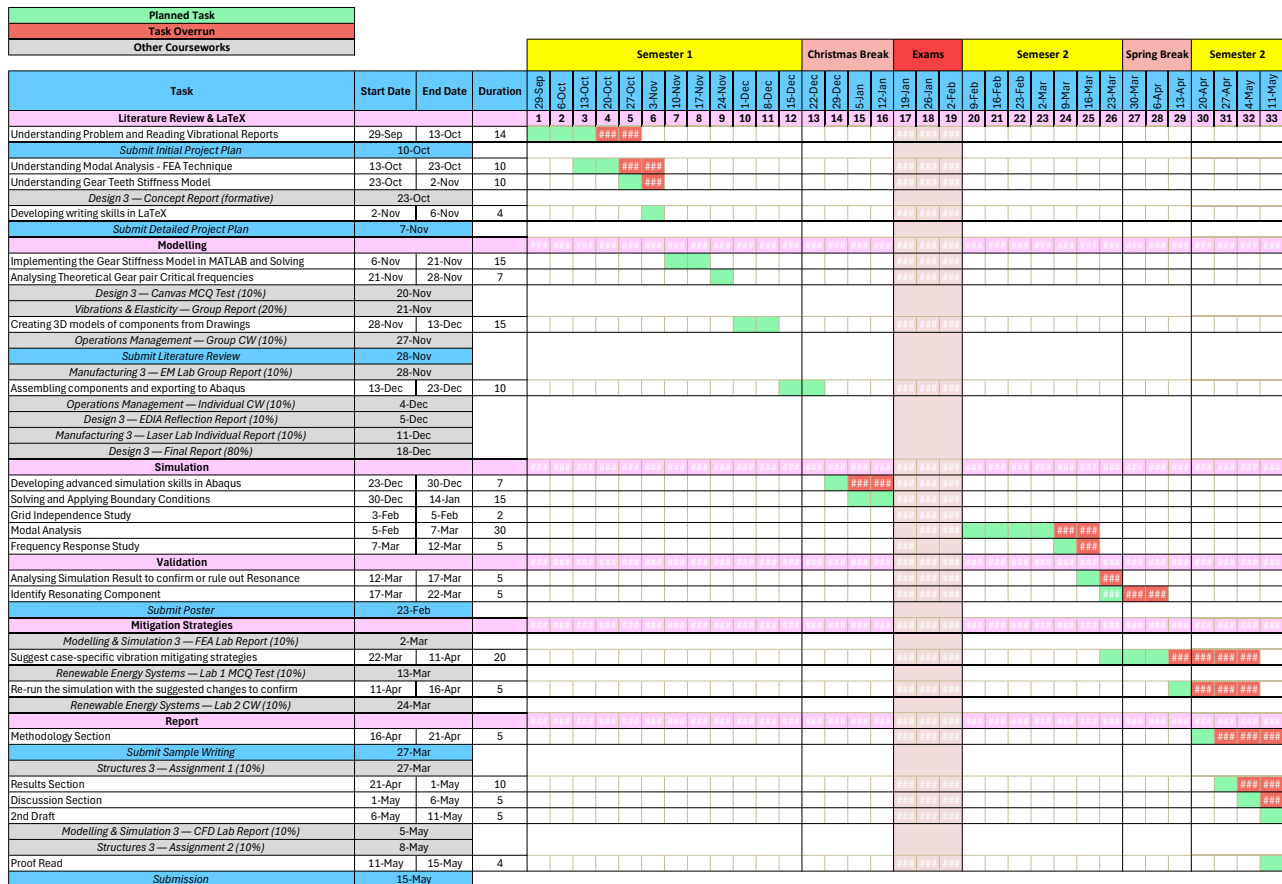


Figure A.4: Final Gantt Chart (May 2026)

B. Appendix B

Risk Matrix						
LIKELIHOOD ↓	IMPACT →					
	Insignificant 1	Minor 2	Significant 3	Major 4	Severe 5	
	Rare 1	LOW 1	LOW 2	LOW 3	MEDIUM 4	MEDIUM 5
	Unlikely 2	LOW 2	MEDIUM 4	MEDIUM 6	MEDIUM 8	HIGH 10
	Moderate 3	LOW 3	MEDIUM 6	HIGH 9	HIGH 12	HIGH 15
	Likely 4	MEDIUM 4	MEDIUM 8	HIGH 12	VERY HIGH 16	VERY HIGH 20
Almost Certain 5	MEDIUM 5	HIGH 10	HIGH 15	VERY HIGH 20	VERY HIGH 25	

Figure B.1: Risk Matrix Scores

B.1.2 — Health & Safety Risk Assessment							
ID	Hazard	Category	L	I	Score	Mitigation	Contingency
H1	Musculoskeletal disorder (back, neck, shoulder) from prolonged sitting	Ergonomic	4	3	12	Height-adjustable chair; monitor at eye level; complete UoM DSE self-assessment; 5-min posture break every 30 min	GP / UoM Occupational Health if symptoms persist; physiotherapy referral; modified working pattern
H2	Eye strain and visual fatigue from prolonged screen use	Ergonomic	5	2	10	20–20–20 rule (every 20 min, focus 20 ft away for 20 s); appropriate screen brightness; blue-light filter during evening sessions	Free eye-test entitlement via UoM for DSE users; reduce daily screen hours
H3	Repetitive strain injury (RSI) of wrists and hands	Ergonomic	3	3	9	Ergonomic keyboard / mouse; wrist-neutral posture; periodic switching between trackpad and mouse; wrist rests	GP referral via UoM Wellbeing; voice-to-text software for the writing phase if needed
H4	Work-related mental stress / anxiety from deadline pressure	Wellbeing	4	3	12	Realistic Gantt with buffer (Section 3.2); weekly progress review with supervisor; maintain hobbies and social contact	UoM Counselling Service (free, confidential); deadline-extension request via Mitigating Circumstances if required
H5	Sleep deprivation from late-night working	Wellbeing	3	3	9	Hard stop at 23:00; no caffeine after 16:00; consistent wake time; do not work in the bedroom	GP for persistent insomnia; reshape schedule to morning-heavy hours
H6	Headache (tension, screen-induced, dehydration)	Wellbeing	3	2	6	2 L water/day target; regular breaks; lower screen brightness in dim rooms	Over-the-counter analgesics; GP if recurrent; rest day taken if severe
H7	Sedentary-lifestyle effects (weight gain, cardiovascular deconditioning)	Wellbeing	4	2	8	30 min daily moderate exercise; walking commute to campus; standing breaks during long runs	UoM Sport facilities membership; book gym time into Gantt as a recurring fixture
H8	Trip hazard at home workstation (cables, clutter, loose rugs)	Workplace	2	2	4	Cable management with clips and sleeves; clear walking routes; adequate floor space around the desk	Domestic first-aid kit; report serious slip to landlord / GP
H9	Electrical hazard — laptop charger, multi-plug overload at home	Workplace	1	4	4	Surge-protected extension lead, no daisy-chaining; weekly visual check of cables; do not work near liquids	Disconnect mains on any sign of damage / heating; PAT testing for any university-loaned equipment
H10	Fire hazard — laptop overheating during long modal runs / charger fire	Workplace	1	5	5	Adequate ventilation; do not run simulations on a bed / cushioned surface; thermal check during overnight runs	Smoke alarm tested monthly; fire blanket accessible; evacuate and dial 999 if ignition occurs
H11	Lone working — late hours in the library / Pariser Building	Workplace	3	2	6	Use well-lit, staffed areas; notify housemate of expected return; UoM SafeZone app active	UoM Security on 0161 306 9966; escort service if available; share live location with a trusted contact
H12	Travel risk — commuting in adverse weather or after dark	External	3	3	9	Use main, well-lit routes; check weather forecast; reflective gear if cycling; carry phone & contactless travel card	Work from home (full remote stack via CSF / RDP available); UoM stay-home provision in extreme weather
H13	Manual handling — carrying laptop, external monitor, and books between locations	Workplace	3	2	6	Padded-strap rucksack; split heavy loads; lift with knees, not back; pause if uncomfortable	Wheeled bag for monitor on transfer days; report any injury via the UoM accident form
H14	Aggravation of a pre-existing condition (e.g., asthma, RSI history, anxiety)	Personal health	2	3	6	Disclose to UoM Disability Advisory & Support Service where applicable; ergonomic setup adapted; medication compliance maintained	Workload adjustment via supervisor; medical extension via Mitigating Circumstances if a flare-up occurs

B.1.1 — General Project Risk Assessment							
ID	Risk	Category	L	I	Score	Mitigation	Contingency
R1	Abaqus licence server unavailable or HE-licence revoked mid-project	Software	2	4	8	UoM HE-licence install on personal laptop; book Pariser CAD-lab PCs as a fallback; pre-cache model files locally	Migrate to Ansys Mechanical (UoM-licensed); supervisor-approved scope reduction to a single sub-assembly
R2	SolidWorks → Abaqus geometry transfer fails (broken faces, mis-stitched surfaces)	Software	3	2	6	Export via neutral STEP/IGES; test workflow on a single bracket before full assembly; check tolerances	Manual geometry repair in Abaqus CAE; rebuild from scratch using simplified primitives
R3	Required MATLAB toolbox (Symbolic / Signal Processing) not in UoM licence pool	Software	1	2	2	Confirm toolbox coverage with IT Services in Week 1; verify each function early	Re-implement with NumPy / SciPy / SymPy in Python (free, open-source)
R4	Software version updates break legacy model files mid-project	Software	1	3	3	Lock to one major version; weekly archive of .cae and .sldprt snapshots	Retain offline installer of the locked version; restore from archive
R5	Excessive simulation runtime (>24 h per modal run)	Modelling	4	4	16	Symmetry exploitation; sub-modelling of contact zones; mesh-size sweep starting coarse; reduce frequency band of interest	Request Computational Shared Facility (CSF) access via supervisor; descope analysis to the gear pair only
R6	Eigenvalue solver fails to converge for the full drive-train assembly	Simulation	3	4	12	Lanczos solver with shift; isolate sub-components; check element distortion; verify connector stiffnesses	Switch to AMS or subspace iteration; analyse pinion and girth gear independently and superpose
R7	Mesh quality issues distort the first non-zero natural frequencies	Simulation	3	3	9	Grid-independence study (already scheduled, Week 14); aspect-ratio and Jacobian checks at gear teeth	Local refinement around contact regions; switch to hex-dominant mesh in critical bodies
R8	Boundary-condition uncertainty (bearing stiffness, foundation compliance) drives unrealistic modes	Modelling	4	3	12	Sensitivity sweep on bearing-stiffness range from literature; conservative envelope per Rao (2017)	Report results as frequency bands rather than point values; document assumptions
R9	Incomplete UltraTech drive-train data / drawings	Data	4	4	16	Cross-reference inspection reports, spare-part datasheets, and OEM catalogues; engineering estimates per ISO 6336 typical sections	Use anonymised generic kiln-drive parameters from FLSmidth / KHD published case studies
R10	Vibrational reports confidential or commercially restricted	Data	3	3	9	Anonymise to RPM ranges and amplitude bands; abstract away plant identifiers	Substitute with peer-reviewed kiln-resonance case studies; declare data limitations in Discussion
R11	ISO 6336 gear-tooth stiffness incorrectly implemented in MATLAB	Modelling	2	4	8	Verify against worked examples (Pedrero, 2010; Lin & Liou, 2014); unit-test each sub-function	Cross-check against KISSsoft trial; request supervisor sanity-check before downstream use
R12	Model over-simplification omits a physically critical mode	Modelling	3	3	9	Each simplification logged with engineering justification; sensitivity test against full model on a single sub-assembly	Re-introduce dropped detail at the validation stage; explicitly bound conclusions
R13	Steep Abaqus learning curve eats into simulation time	Knowledge	5	2	10	7 days budgeted in Gantt; Dassault tutorials; reuse code from UoM module materials	Pairing session with a PhD student in the supervisor's group; targeted supervisor demo
R14	Steep LaTeX learning curve delays writing	Knowledge	4	1	4	5 days budgeted; Overleaf templates; learn-as-you-write approach with minimal early commands	Fall back to Word + Mendeley for final two weeks if blocked (last resort)
R15	Limited literature on rotary-kiln gear-mesh resonance specifically	Literature	3	2	6	Broaden search to generic rotor dynamics, gearbox NVH, planetary-gear vibration; supervisor reading list	Adapt methodology from analogous cases (mining mills, ship gearboxes); declare scope in Introduction
R16	Sem 1 & Sem 2 coursework deadline clashes	Time	5	3	15	Integrated Gantt covering every module deadline (see Section 3.2); buffer days pre-allocated; weekly status check	Request short coursework extensions where eligible; flag clashes to year tutor early
R17	Personal commitments (placement / graduate-school applications) eat into project time	Personal	4	2	8	Diary-block all applications; communicate upcoming peaks to supervisor; pre-deadline buffer	Re-prioritise scope; defer non-critical extensions to "Further Work" in the report
R18	Christmas / Easter campus closures disrupt continuity	External	4	2	8	Local Abaqus install for at-home work; reading list pre-cached; offline copies of all data	Use breaks for non-software tasks (writing, sketching, figure preparation)
R19	File corruption / accidental data loss	File mgmt.	2	4	8	GitHub repository for code & LaTeX sources; OneDrive auto-sync; weekly external-HDD backup	Restore from latest cloud snapshot; recover via Git commit history
R20	Personal laptop hardware failure	File mgmt.	2	4	8	Cloud-only working files; remote access via CSF/RDP; no project files stored only on local disk	UoM lab PCs as primary; emergency laptop loan from IT Services
R21	Inconclusive results — resonance is not the cause of observed vibrations	Methodology	3	3	9	Negative results documented; alternate causes discussed (foundation, alignment, casting defects)	Re-frame report as elimination study; supervisor-approved scope shift to mode-ranking only
R22	Methodology flaw discovered late in the project	Methodology	2	5	10	Early validation against benchmark problems (cantilever beam, simple rotor); mid-project supervisor checkpoint	Buffer in the 2nd-Draft phase of the Gantt absorbs late corrections; descope if needed
R23	Supervisor unavailable for an extended period (illness, conference)	Communication	2	4	8	Recurring weekly meetings booked at the start of term; written queries logged via email	Approach year tutor or co-supervisor; consult PhD students in the supervisor's research group
R24	Scope creep beyond the agreed objectives	Scope	3	3	9	Objectives fixed in the Project Plan (Section 1.2); supervisor sign-off required for additions	Defer extra-objective work to "Future Work" in the Discussion section
R25	Plagiarism or referencing error / academic-malpractice flag	Academic integrity	1	5	5	Mendeley citation manager from day 1; Turnitin self-check before submission; explicit attribution of all figures and code	Address via UoM Academic Malpractice Procedure if flagged; immediate supervisor disclosure